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United States Patent	12412854
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Chen; Linghan et al.

Semiconductor devices containing copper bonding pads with different conductive barrier layers and methods for forming the same

Abstract

Bonding strength and yield can be enhanced by providing a mating pair of a convex bonding surface and a concave bonding surface. The convex bonding surface can be provided by employing a conductive barrier layer having a higher electrochemical potential than copper. The concave bonding surface can be provided by employing a conductive barrier layer having a lower electrochemical potential than copper. Alternatively additionally, a copper material portion in a bonding pad may include at least 10% volume fraction of (200) copper grains to provide high volume expansion toward a mating copper material portion. The mating copper material portion may be formed with at least 95% volume fraction of (111) copper grains to provide high surface diffusivity, or may be formed with at least 10% volume fraction of (200) copper grains to provide high volume expansion.

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Appl. No.: **17/662493**

Filed: **May 09, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230361066 A1	Nov. 09, 2023

Publication Classification

Int. Cl.: **H01L23/00** (20060101); **H01L25/00** (20060101); **H01L25/065** (20230101); **H01L25/18** (20230101)

U.S. Cl.:

CPC **H01L24/08** (20130101); **H01L24/03** (20130101); **H01L24/05** (20130101); **H01L24/80** (20130101); **H01L25/0657** (20130101); **H01L25/18** (20130101); **H01L25/50** (20130101); H01L2224/03616 (20130101); H01L2224/05082 (20130101); H01L2224/05147 (20130101); H01L2224/05157 (20130101); H01L2224/05166 (20130101); H01L2224/05176 (20130101); H01L2224/05181 (20130101); H01L2224/0807 (20130101); H01L2224/08145 (20130101); H01L2224/80895 (20130101); H01L2224/80896 (20130101); H01L2924/04941 (20130101); H01L2924/04953 (20130101); H01L2924/1431 (20130101); H01L2924/14511 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

FIELD

(1) The present disclosure relates generally to the field of semiconductor devices, and particularly to a semiconductor structure including a bonded assembly of semiconductor dies that are bonded via copper-to-copper bonding and methods for forming the same.

BACKGROUND

(2) Process reliability and yield of copper-to-copper bonding between mating pairs of copper pads

becomes more challenging as the dimensions of the copper pads decrease in advanced semiconductor technology generations. Reducing volumes of cavities between mating pairs of copper pads is desirable to improve copper-to-copper bonding during chip bonding.

SUMMARY

(3) According to an aspect of the present disclosure, a bonded assembly comprises a first semiconductor die comprising first semiconductor devices and a first bonding pad, wherein the first bonding pad comprises a first copper material portion and a first conductive barrier layer comprising a first conductive barrier material having a higher electrochemical potential than copper located between the first semiconductor devices and the first copper material portion, and a second semiconductor die comprising second semiconductor devices and a second bonding pad, wherein the second bonding pad comprises a second copper material portion and a second conductive barrier layer comprising a second conductive barrier material having a lower electrochemical potential than copper located between the second semiconductor devices and the second copper material portion. The second bonding pad is bonded to the first bonding pad.

(4) According to another aspect of the present disclosure, a method of forming a bonded assembly is provided, which comprises: providing a first semiconductor die comprising first semiconductor devices, first metal interconnect structures embedded in first dielectric material layers and electrically connected to the first semiconductor devices, and a first bonding pad embedded in a first bonding-level dielectric layer and electrically connected to one of the first metal interconnect structures, wherein the first bonding pad comprises a first conductive barrier layer comprising a first conductive barrier material having a higher electrochemical potential than copper and a first copper material portion at least partially laterally surrounded by the first conductive barrier layer; providing a second semiconductor die comprising second semiconductor devices, second metal interconnect structures embedded in second dielectric material layers and electrically connected to the second semiconductor devices, and a second bonding pad embedded in a second bonding-level dielectric layer and electrically connected to one of the second metal interconnect structures, wherein the second bonding pad comprises a second conductive barrier layer comprising a second conductive barrier material having a lower electrochemical potential than copper and a second copper material portion at least partially laterally surrounded by the second conductive barrier layer; and bonding the second bonding pad to the first bonding pad by inducing copper-to-copper bonding between the second copper material portion and the first copper material portion.

(5) According to an aspect of the present disclosure, a bonded assembly comprises a first semiconductor die comprising first semiconductor devices and a first bonding pad, wherein the first bonding pad comprises a first copper material portion containing (200) copper grains at a volume fraction of at least 10% and a first conductive barrier layer located between the first semiconductor devices and the first copper material portion, and a second semiconductor die comprising second semiconductor devices and a second bonding pad, wherein the second bonding pad comprises a second copper material portion and a second conductive barrier layer located between the second semiconductor devices and the second copper material portion. The second bonding pad is bonded to the first bonding pad.

(6) According to another aspect of the present disclosure, a bonded assembly is provided, which comprises: a first semiconductor die comprising first semiconductor devices, first metal interconnect structures embedded in first dielectric material layers and electrically connected to the first semiconductor devices, and a first bonding pad embedded in a first bonding-level dielectric layer and electrically connected to one of the first metal interconnect structures, wherein the first bonding pad comprises a first conductive barrier layer and a first copper material portion at least partially laterally surrounded by the first conductive barrier layer and including (200) copper grains at a volume fraction of at least 10%; and a second semiconductor die comprising second semiconductor devices, second metal interconnect structures embedded in second dielectric material layers and electrically connected to the second semiconductor devices, and a second

bonding pad embedded in a second bonding-level dielectric layer and electrically connected to one of the second metal interconnect structures, wherein the second bonding pad comprises a second conductive barrier layer and a second copper material portion at least partially laterally surrounded by the second conductive barrier layer and including (200) copper grains at a volume fraction not less than 10%, wherein the second bonding pad is bonded to the first bonding pad.
(7) According to yet another aspect of the present disclosure, a method of forming a bonded assembly is provided, which comprises: providing a first semiconductor die comprising first semiconductor devices, first metal interconnect structures embedded in first dielectric material layers and electrically connected to the first semiconductor devices, and a first bonding pad embedded in a first bonding-level dielectric layer and electrically connected to one of the first metal interconnect structures, wherein the first bonding pad comprises a first conductive barrier layer and a first copper material portion laterally surrounded by the first conductive barrier layer and including (200) copper grains at a volume fraction not less than 10%; providing a second semiconductor die comprising second semiconductor devices, second metal interconnect structures embedded in second dielectric material layers and electrically connected to the second semiconductor devices, and a second bonding pad embedded in a second bonding-level dielectric layer and electrically connected to one of the second metal interconnect structures, wherein the second bonding pad comprises a second conductive barrier layer and a second copper material portion laterally surrounded by the second conductive barrier layer; and bonding the second bonding pad is bonded to the first bonding pad by inducing copper-to-copper bonding between the second copper material portion and the first copper material portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a schematic vertical cross-sectional view of a first configuration of a memory die after formation of memory-side pad cavities according to a first embodiment of the present disclosure.
(2) FIG. **1**B is a schematic vertical cross-sectional view of the first configuration of the memory die after formation of a memory-side conductive barrier layer and a memory-side copper layer according to the first embodiment of the present disclosure.
(3) FIG. **1**C is a schematic vertical cross-sectional view of the first configuration of the memory die along plane C-C′ in FIG. **1**D after formation of memory-side bonding pads according to the first embodiment of the present disclosure.
(4) FIG. **1**D is a schematic top-down view of the first configuration of the memory die after formation of the memory-side bonding pads according to the first embodiment of the present disclosure.
(5) FIG. **2**A is a schematic vertical cross-sectional view of a second configuration of a logic die after formation of logic-side pad cavities according to the first embodiment of the present disclosure.
(6) FIG. **2**B is a schematic vertical cross-sectional view of the second configuration of the logic die after formation of a logic-side conductive barrier layer and a logic-side copper layer according to the first embodiment of the present disclosure.
(7) FIG. **2**C is a schematic vertical cross-sectional view of the second configuration of the logic die after formation of logic-side bonding pads according to the first embodiment of the present disclosure.
(8) FIG. **2**D is a schematic top-down view of the second configuration of the logic die after formation of the logic-side bonding pads according to the first embodiment of the present disclosure.

(9) FIG. **3**A is a schematic vertical cross-sectional view of a first configuration of a first exemplary structure after the logic die is disposed on the memory die for bonding according to the first embodiment of the present disclosure.
(10) FIG. **3**B is a schematic vertical cross-sectional view of the first configuration of the first exemplary structure after bonding the logic die to the memory die according to the first embodiment of the present disclosure.
(11) FIGS. **3**C and **3**D are schematic vertical cross-sectional views of alternative first configurations of the first exemplary structure after thinning the memory die according to the first embodiment of the present disclosure.
(12) FIG. **4**A is a schematic vertical cross-sectional view of a second configuration of the first exemplary structure after the logic die is disposed on the memory die for bonding according to the first embodiment of the present disclosure.
(13) FIG. **4**B is a schematic vertical cross-sectional view of the second configuration of the first exemplary structure after bonding the logic die to the memory die according to the first embodiment of the present disclosure.
(14) FIGS. **4**C and **4**D are schematic vertical cross-sectional views of alternative second configurations of the first exemplary structure after thinning the memory die according to the first embodiment of the present disclosure.
(15) FIG. **5** is a schematic vertical cross-sectional view of a third configuration of a logic die after formation of logic-side bonding pads according to the first embodiment of the present disclosure.
(16) FIG. **6**A is a schematic vertical cross-sectional view of the third configuration of the first exemplary structure after bonding the logic die to the memory die according to the first embodiment of the present disclosure.
(17) FIG. **6**B is a schematic vertical cross-sectional view of the third configuration of the first exemplary structure after thinning the memory die and formation of external bonding structures according to the first embodiment of the present disclosure.
(18) FIG. **7**A is a schematic vertical cross-sectional view of a fourth configuration of the first exemplary structure after bonding the logic die to the memory die according to the first embodiment of the present disclosure.
(19) FIG. **7**B is a schematic vertical cross-sectional view of the fourth configuration of the first exemplary structure after thinning the memory die and formation of external bonding structures according to the first embodiment of the present disclosure.
(20) FIG. **8**A is a vertical cross-sectional view of a first configuration of a memory die after formation of a memory-side conductive barrier layer and a memory-side copper layer according to a second embodiment of the present disclosure.
(21) FIG. **8**B is a vertical cross-sectional view of the first configuration of the memory die after formation of a memory-side bonding pads according to a second embodiment of the present disclosure.
(22) FIG. **9**A is a vertical cross-sectional view of a first configuration of a logic die after formation of a logic-side conductive barrier layer and a logic-side copper layer according to a second embodiment of the present disclosure.
(23) FIG. **9**B is a vertical cross-sectional view of the first configuration of the logic die after formation of a logic-side bonding pads according to a second embodiment of the present disclosure.
(24) FIG. **10**A is a schematic vertical cross-sectional view of a first configuration of a second exemplary structure after the logic die is disposed on the memory die for bonding according to the second embodiment of the present disclosure.
(25) FIG. **10**B is a schematic vertical cross-sectional view of the first configuration of the second exemplary structure after bonding the logic die to the memory die according to the second embodiment of the present disclosure.

(26) FIGS. **10**C and **10**D are schematic vertical cross-sectional views of alternative first configurations of the second exemplary structure after thinning the memory die according to the second embodiment of the present disclosure.
(27) FIG. **11**A is a vertical cross-sectional view of a second configuration of a logic die after formation of a logic-side conductive barrier layer and a logic-side copper layer according to a second embodiment of the present disclosure.
(28) FIG. **11**B is a vertical cross-sectional view of the second configuration of the logic die after formation of a logic-side bonding pads according to a second embodiment of the present disclosure.
(29) FIG. **12**A is a schematic vertical cross-sectional view of a second configuration of a second exemplary structure after the logic die is disposed on the memory die for bonding according to the second embodiment of the present disclosure.
(30) FIG. **12**B is a schematic vertical cross-sectional view of the second configuration of the second exemplary structure after bonding the logic die to the memory die according to the second embodiment of the present disclosure.
(31) FIG. **12**C is a schematic vertical cross-sectional view of the second configuration of the second exemplary structure after thinning the memory die and formation of external bonding structures according to the second embodiment of the present disclosure.
(32) FIG. **13**A is a vertical cross-sectional view of a third configuration of a memory die after formation of a memory-side conductive barrier layer and a memory-side copper layer according to a second embodiment of the present disclosure.
(33) FIG. **13**B is a vertical cross-sectional view of the third configuration of the memory die after formation of a memory-side bonding pads according to a second embodiment of the present disclosure.
(34) FIG. **14**A is a schematic vertical cross-sectional view of a third configuration of a second exemplary structure after the logic die is disposed on the memory die for bonding according to the second embodiment of the present disclosure.
(35) FIG. **14**B is a schematic vertical cross-sectional view of the third configuration of the second exemplary structure after bonding the logic die to the memory die according to the second embodiment of the present disclosure.
(36) FIG. **14**C is a schematic vertical cross-sectional view of the third configuration of the second exemplary structure after thinning the memory die and formation of external bonding structures according to the second embodiment of the present disclosure.

DETAILED DESCRIPTION

(37) The embodiments of the present disclosure are directed to a semiconductor structure including a bonded assembly of semiconductor dies that are bonded via copper-to-copper bonding and methods for forming the same. The different conductive barrier layers may be used in various embodiments to increase copper bonding yield for the bonded. The various aspects of embodiments of the present disclosure are described in detail herebelow.
(38) The drawings are not drawn to scale. Multiple instances of an element may be duplicated where a single instance of the element is illustrated, unless absence of duplication of elements is expressly described or clearly indicated otherwise. Ordinals such as “first,” “second,” and “third” are used merely to identify similar elements, and different ordinals may be used across the specification and the claims of the instant disclosure. The term “at least one” element refers to all possibilities including the possibility of a single element and the possibility of multiple elements.
(39) The same reference numerals refer to the same element or similar element. Unless otherwise indicated, elements having the same reference numerals are presumed to have the same composition and the same function. Unless otherwise indicated, a “contact” between elements refers to a direct contact between elements that provides an edge or a surface shared by the elements. If two or more elements are not in direct contact with each other or among one another,

the two elements are “disjoined from” each other or “disjoined among” one another. As used herein, a first element located “on” a second element can be located on the exterior side of a surface of the second element or on the interior side of the second element. As used herein, a first element is located “directly on” a second element if there exist a physical contact between a surface of the first element and a surface of the second element. As used herein, a first element is “electrically connected to” a second element if there exists a conductive path consisting of at least one conductive material between the first element and the second element. As used herein, a “prototype” structure or an “in-process” structure refers to a transient structure that is subsequently modified in the shape or composition of at least one component therein.

(40) As used herein, a “layer” refers to a material portion including a region having a thickness. A layer may extend over the entirety of an underlying or overlying structure, or may have an extent less than the extent of an underlying or overlying structure. Further, a layer may be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer may be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer may extend horizontally, vertically, and/or along a tapered surface. A substrate may be a layer, may include one or more layers therein, or may have one or more layer thereupon, thereabove, and/or therebelow.

(41) As used herein, a first surface and a second surface are “vertically coincident” with each other if the second surface overlies or underlies the first surface and there exists a vertical plane or a substantially vertical plane that includes the first surface and the second surface. A substantially vertical plane is a plane that extends straight along a direction that deviates from a vertical direction by an angle less than 5 degrees. A vertical plane or a substantially vertical plane is straight along a vertical direction or a substantially vertical direction, and may, or may not, include a curvature along a direction that is perpendicular to the vertical direction or the substantially vertical direction.

(42) As used herein, a “memory level” or a “memory array level” refers to the level corresponding to a general region between a first horizontal plane (i.e., a plane parallel to the top surface of the substrate) including topmost surfaces of an array of memory elements and a second horizontal plane including bottommost surfaces of the array of memory elements. As used herein, a “through-stack” element refers to an element that vertically extends through a memory level.

(43) As used herein, a “semiconducting material” refers to a material having electrical conductivity in the range from 1.0×10.sup.−5 S/m to 1.0×10.sup.5 S/m. As used herein, a “semiconductor material” refers to a material having electrical conductivity in the range from 1.0×10.sup.−5 S/m to 1.0 S/m in the absence of electrical dopants therein, and is capable of producing a doped material having electrical conductivity in a range from 1.0 S/m to 1.0×10.sup.5 S/m upon suitable doping with an electrical dopant. As used herein, an “electrical dopant” refers to a p-type dopant that adds a hole to a valence band within a band structure, or an n-type dopant that adds an electron to a conduction band within a band structure. As used herein, a “conductive material” refers to a material having electrical conductivity greater than 1.0×10.sup.5 S/m. As used herein, an “insulator material” or a “dielectric material” refers to a material having electrical conductivity less than 1.0×10.sup.−5 S/m. As used herein, a “heavily doped semiconductor material” refers to a semiconductor material that is doped with electrical dopant at a sufficiently high atomic concentration to become a conductive material either as formed as a crystalline material or if converted into a crystalline material through an anneal process (for example, from an initial amorphous state), i.e., to have electrical conductivity greater than 1.0×10.sup.5 S/m. A “doped semiconductor material” may be a heavily doped semiconductor material, or may be a semiconductor material that includes electrical dopants (i.e., p-type dopants and/or n-type dopants) at a concentration that provides electrical conductivity in the range from 1.0×10.sup.−5 S/m to 1.0×10.sup.5 S/m. An “intrinsic semiconductor material” refers to a semiconductor material that is not doped with electrical dopants. Thus, a semiconductor material may be semiconducting or conductive, and may be an intrinsic semiconductor material or a doped semiconductor material. A

doped semiconductor material may be semiconducting or conductive depending on the atomic concentration of electrical dopants therein. As used herein, a “metallic material” refers to a conductive material including at least one metallic element therein. All measurements for electrical conductivities are made at the standard condition.

(44) Generally, a semiconductor package (or a “package”) refers to a unit semiconductor device that may be attached to a circuit board through a set of pins or solder balls. A semiconductor package may include a semiconductor chip (or a “chip”) or a plurality of semiconductor chips that are bonded throughout, for example, by flip-chip bonding or another chip-to-chip bonding. A package or a chip may include a single semiconductor die (or a “die”) or a plurality of semiconductor dies. A die is the smallest unit that may independently execute external commands or report status. Typically, a package or a chip with multiple dies is capable of simultaneously executing as many external commands as the total number of planes therein. Each die includes one or more planes. Identical concurrent operations may be executed in each plane within a same die, although there may be some restrictions. In case a die is a memory die, i.e., a die including memory elements, concurrent read operations, concurrent write operations, or concurrent erase operations may be performed in each plane within a same memory die. In a memory die, each plane contains a number of memory blocks (or “blocks”), which are the smallest unit that may be erased by in a single erase operation. Each memory block contains a number of pages, which are the smallest units that may be selected for programming. A page is also the smallest unit that may be selected to a read operation.

(45) Referring to FIG. **1**A, a memory die **900** is illustrated. First-type bonding pads, which are also referred to as first bonding pads, can be subsequently formed on the memory die **900**. In this case, the memory die **900** can be referred to as a first-type semiconductor die, or as a first semiconductor die **100**. The memory die **900** includes a memory-side substrate **908**, memory-side semiconductor devices **920** overlying the memory-side substrate **908**, memory-side dielectric material layers (**290**, **950**) located on the memory-side semiconductor devices, and memory-side metal interconnect structures **960** embedded in the memory-side dielectric material layers (**290**, **950**). In one embodiment, the memory-side substrate **908** may be a portion of a commercially available silicon wafer having a thickness in a range from 500 microns to 1 mm. In one embodiment, a plurality of first semiconductor dies **100** can be provided on the memory-side substrate **908**, which may be a single crystal silicon wafer, or another suitable substrate.

(46) Optional discrete substrate recess cavities can be formed in an upper portion of the memory-side substrate **908** by applying a photoresist layer over the top surface of the memory-side substrate **908**, lithographically patterning the photoresist layer to form an array of discrete openings, and transferring the pattern of the array of discrete openings into the upper portion of the memory-side substrate by performing an anisotropic etch process. The photoresist layer can be subsequently removed, for example, by ashing. The depth of each discrete substrate recess cavity can be in a range from 500 nm to 10,000, although lesser and greater depths can also be employed. Optional through-substrate liner **386** and optional through-substrate via structure **388** can be formed within each discrete substrate recess cavity.

(47) Generally, the memory-side semiconductor devices **920** may comprise any semiconductor device known in the art. In one embodiment, the memory die **900** comprises a memory die, and may include memory devices, such as a three-dimensional NAND memory device. In an illustrative example, the memory-side semiconductor devices **920** may include a vertically alternating stack of insulating layers **32** and electrically conductive layers **46**, and a two-dimensional array of memory openings vertically extending through the vertically alternating stack (**32**, **46**). The electrically conductive layers **46** may comprise word lines of the three-dimensional NAND memory device.

(48) A memory opening fill structure **58** may be formed within each memory opening. A memory opening fill structure **58** may include a memory film and a vertical semiconductor channel

contacting the memory film. The memory film may include a blocking dielectric, a tunneling dielectric and a charge storage material located between the blocking and tunneling dielectric. The charge storage material may comprise charge trapping layer, such as a silicon nitride layer, or a plurality of discrete charge trapping regions, such as floating gates or discrete portions of a charge trapping layer. In this case, each memory opening fill structure **58** and adjacent portions of the electrically conductive layers **46** constitute a vertical NAND string. Alternatively, the memory opening fill structures **58** may include any type of non-volatile memory elements such as resistive memory elements, ferroelectric memory elements, phase change memory elements, etc. The memory device may include an optional horizontal semiconductor channel layer **10** connected to the bottom end of each vertical semiconductor channel, and an optional dielectric spacer layer **910** that provides electrical isolation between the memory-side substrate **908** and the horizontal semiconductor channel layer **10**.

(49) The electrically conductive layers **46** may be patterned to provide a terrace region in which each overlying electrically conductive layer **46** has a lesser lateral extent than any underlying electrically conductive layer **46**. Contact via structures (not shown) may be formed on the electrically conductive layers **46** in the terrace region to provide electrical connection to the electrically conductive layers **46**. Dielectric material portions **65** may be formed around each vertically alternating stack (**32**, **46**) to provide electrical isolation between neighboring vertically alternating stacks (**32**, **46**).

(50) Through-memory-level via cavities can be formed through the dielectric material portions **65**, the optional dielectric spacer layer **910**, and the horizontal semiconductor channel layer **10**. An optional through-memory-level dielectric liner **486** and a through-memory-level via structure **488** can be formed within each through-memory-level via cavity. Each through-memory-level dielectric liner **486** includes a dielectric material such as silicon oxide. Each through-memory-level via structure **488** can be formed directly on a respective one of the through-substrate via structure **388**.

(51) The memory-side dielectric material layers (**290**, **950**) may include memory-side proximal dielectric material layers **290** embedding contact via structures and bit lines **962** and memory-side distal dielectric material layers **950** that embed a subset of the memory-side metal interconnect structures **960** located above the memory-side proximal dielectric material layers **290**. As used herein, a “proximal” surface refers to a surface that is close to a substrate, and a “distal” surface refers to a surface that is distal from the substrate. In the memory die **900**, a proximal surface refers to a surface that is close to the memory-side substrate **908**, and a distal surface refers to a surface that is distal from the memory-side substrate **908**.

(52) The bit lines **962** may electrically contact drain regions located above the semiconductor channel at the top of the memory opening fill structures **58**. The contact via structures contact various nodes of the memory-side semiconductor devices, such as the electrically conductive layers **46** (e.g., word lines and select gate electrodes). Generally, the memory-side metal interconnect structures **960** can be electrically connected to the memory-side semiconductor devices **920**.

(53) Each of the memory-side proximal dielectric material layers **290** and the memory-side distal dielectric material layers **950** may include a dielectric material such as undoped silicate glass, a doped silicate glass, organosilicate glass, silicon nitride, a dielectric metal oxide, or a combination thereof. The memory-side distal dielectric material layers **950** may include one or more dielectric diffusion barrier layers (not expressly shown). In this case, each dielectric diffusion barrier layer embedded in the memory-side distal dielectric material layers **950** may include silicon carbon nitride (i.e., silicon carbonitride “SiCN”, which is also referred to silicon carbide nitride), silicon nitride (Si.sub.3N.sub.4), silicon oxynitride, or any other dielectric material that is effective in blocking diffusion of copper. In one embodiment, each dielectric diffusion barrier layer embedded in the memory-side distal dielectric material layers **950** may include a dielectric material having a dielectric constant less than 5, such as SiCN having a dielectric constant of about 3.8, to reduce RC delay of the memory-side metal interconnect structures **960**. Each dielectric diffusion barrier layer

may have a thickness in a range from 10 nm to 30 nm.
(54) An optional memory-side dielectric diffusion barrier layer **952** can be formed over the memory-side dielectric material layers (**290**, **950**). The memory-side dielectric diffusion barrier layer **952** can include a dielectric material that blocks copper diffusion. In one embodiment, the memory-side dielectric diffusion barrier layer **952** can include silicon nitride, silicon carbon nitride, silicon oxynitride, or a stack thereof. The thickness of the memory-side dielectric diffusion barrier layer **952** can be in a range from 5 nm to 50 nm, although lesser and greater thicknesses can also be employed.
(55) A memory-side bonding-level dielectric layer **970** can be formed over the memory-side dielectric diffusion barrier layer **952**. In case first bonding pads are subsequently formed in the memory-side bonding-level dielectric layer **970**, the memory-side bonding-level dielectric layer **970** is herein referred to as a first bonding-level dielectric layer. Optionally, the memory-side bonding-level dielectric layer **970** may include a dielectric material that permits dielectric-to-dielectric bonding. The memory-side bonding-level dielectric layer **970** may include, and/or consist essentially of, undoped silicate glass (i.e., silicon oxide), a doped silicate glass, organosilicate glass, silicon nitride, or a dielectric metal oxide. The thickness of the memory-side bonding-level dielectric layer **970** may be in a range from 300 nm to 3,000 nm, although lesser and greater thicknesses may also be employed. The memory-side bonding-level dielectric layer **970** may have a planar top surface.
(56) A photoresist layer (not shown) can be applied over the memory-side bonding-level dielectric layer **970**, and can be lithographically patterned to form discrete openings in each area in which first bonding pads are to be subsequently formed. In one embodiment, the first bonding pads may have shape similar to a mushroom including a lower via (e.g., stem) portion and an upper plate (e.g., cap) portion having a greater lateral dimension than the lower via portion. In this case, discrete opening in the photoresist layer can have shapes of the lower via portions of the first bonding pads. A first anisotropic etch process can be performed to transfer the pattern of the openings in the photoresist layer into an upper portion of the memory-side bonding-level dielectric layer **970**. Recess cavities can be formed within unmasked areas in the upper portion of the memory-side bonding-level dielectric layer **970**. The photoresist layer may be subsequently isotropically trimmed to increase the size of each opening in the photoresist layer. Alternatively, the photoresist layer may be removed and a new photoresist layer may be applied and patterned over the memory-side bonding-level dielectric layer **970** to form larger openings having peripheries that are laterally offset outward from a periphery of a respective recess cavity in the upper portion of the memory-side bonding-level dielectric layer **970**. A second anisotropic etch process can be performed to etch portions of the memory-side bonding-level dielectric layer **970** that are not covered by the photoresist layer. Memory-side pad cavities **979** are formed through the memory-side bonding-level dielectric layer **970** underneath discrete openings in the photoresist layer. The photoresist layer may be subsequently removed, for example, by ashing.
(57) Yet alternatively, the first bonding pads may be formed with a respective straight sidewall that vertically extends from a top surface of the memory-side bonding-level dielectric layer **970** to a bottom surface of the memory-side bonding-level dielectric layer **970**. In this case, the memory-side pad cavities **979** may be formed by applying and patterning a photoresist layer, and by removing portions of the memory-side bonding-level dielectric layer **970** by performing an anisotropic etch process. In other words, a combination of a single lithographic patterning process and a single anisotropic etch process may be employed to form the memory-side pad cavities **979** with straight sidewall(s). In one embodiment, a lateral dimension (such as a diameter or a lateral distance between facing sides of a periphery) of each opening in the photoresist layer may be in a range from 200 nm to 30 microns, such as from 400 nm to 20 microns, although lesser and greater dimensions may also be employed.
(58) A top surface of a respective memory-side metal interconnect structure **960** can be physically

exposed at the bottom of each memory-side pad cavity **979**. In one embodiment, each memory-side pad cavity **979** can have a top periphery having a horizontal cross-sectional shape of a rectangle or a rounded rectangle. A maximum lateral dimension of each memory-side pad cavity **979** may be in a range from 200 nm to 30 microns, such as from 400 nm to 10 microns, although lesser and greater dimensions may also be employed. Sidewalls of the memory-side pad cavities **979** may be vertical, or may have a taper angle greater than 0 degree and less than 30 degrees (such as a taper angle in a range from 3 degrees to 10 degrees) with respect to the vertical direction. In case first bonding pads (e.g., first-type bonding pads) are subsequently formed in the memory-side pad cavities **979**, the memory-side pad cavities **979** are herein referred to as first pad cavities.

(59) Referring to FIG. **1**B, a memory-side conductive (i.e., electrically conductive) barrier layer **984**L and a memory-side copper layer **986**L can be sequentially deposited in the memory-side pad cavities **979**. In one embodiment, the memory-side conductive barrier layer **984**L may be formed as a first-type conductive barrier layer, which is herein referred to as a first conductive barrier layer **14**L. In this case, the memory-side copper layer **986**L is formed on the first conductive barrier layer **14**L, and is herein referred to as a first copper layer **16**L.

(60) According to an aspect of the present disclosure, the first conductive barrier layer **14**L comprises, and/or consists essentially of, a first conductive barrier material having a higher electrochemical potential than copper. Electrochemical potential is the mechanical work done in bringing 1 mole of an ion from a standard state to a specified concentration and electrical potential. International Union of Pure and Applied Chemistry defines the electrochemical potential as the partial molar Gibbs energy of a substance at the specified electric potential, where the substance is in a specified phase. In the present disclosure, all electric potentials are zero, and all materials are in a solid phase for the purpose of calculation of electrochemical potential. Generally, electrochemical potential can be expressed as:

$$\mu_i=\mu_i+z_i F\Phi,$$

in which μ_i is the electrochemical potential of species i, in J/mol, μ_i is the chemical potential of the species i, in J/mol, z_i is the valency (charge) of the ion i, a dimensionless integer, F is the Faraday constant, in C/mol, Φ is the local electrostatic potential, in V. In the special case of an uncharged atom, $z_i=0$, and so $\mu_i=\mu_i$, which is the case for all materials under discussion in the present disclosure. In other words, the electrochemical potential for all material portions of the present disclosure is the chemical potential of a respective material portion for the purpose of the present disclosure because all material portions are at an equipotential state.

(61) Non-limiting examples of electrically conductive materials that provide higher electrochemical potential than copper and provide suitable adhesion strength to a dielectric material (such as silicon oxide) of the first bonding-level dielectric layer (such as the memory-side bonding-level dielectric layer **970**) comprise cobalt, tantalum, or tantalum nitride. In one embodiment, the first conductive barrier layer **14** consists essentially of a material selected from cobalt, tantalum, or tantalum nitride. The thickness of the first conductive barrier layer **14**L (e.g., the memory-side conductive barrier layer **984**L) may be in a range from 3 nm to 30 nm, such as from 6 nm to 20 nm, although lesser and greater thicknesses can also be employed.

(62) The first copper layer **16**L (e.g., memory-side copper layer **986**L) may consist essentially of copper. The first copper layer **16**L may be formed by depositing a copper seed layer (which is a lower surface portion of the first copper layer **16**L) employing a physical vapor deposition process, and by depositing a copper fill material layer employing electroplating. The thickness of the copper seed layer may be in a range from 10 nm to 100 nm, and the thickness of the copper fill material layer, and measured over the top surface of the first bonding-level dielectric layer, may be in a range from 300 nm to 6,000 nm, although lesser and greater thicknesses may also be employed.

(63) Referring to FIGS. **1**C and **1**D, a first chemical mechanical polishing (CMP) process can be performed to remove portions of the first copper layer **16**L (e.g., the memory-side copper layer **986**L) and the first conductive barrier layer **14**L (e.g., the memory-side conductive barrier layer

984L) from above the top surface of the first bonding-level dielectric layer (e.g., the memory-side bonding-level dielectric layer **970**). Each remaining portion of the first copper layer **16**L comprises the first copper material portion **16**. Each remaining portion of the first conductive barrier layer **14**L comprises a first conductive barrier layer **14**. Each first copper material portion **16** in this embodiment is a memory-side copper material portion **986**. Each first conductive barrier layer **14**L in this embodiment is a memory-side conductive barrier layer **984** embedded within the memory-side bonding dielectric layer **970**. Each contiguous combination of a first conductive barrier layer **14** and a first copper material portion **16** constitutes a first bonding pad **18**, which in this embodiment is a memory-side bonding pad **988**.

(64) Generally, first bonding pads **18** embedded in a first bonding-level dielectric layer (such as the memory-side bonding-level dielectric layer **970**) and electrically connected to a respective one of the first metal interconnect structures (such as the memory-side metal interconnect structures **960**) are formed. Each of the first bonding pads **18** comprises a first conductive barrier layer **14** including, and/or consisting essentially of, a first conductive barrier material having a higher electrochemical potential than copper. Each of the first bonding pads **18** comprises a first copper material portion **16** laterally surrounded by the first conductive barrier layer **14**.

(65) The first chemical mechanical polishing process proceeds under a condition in which the first conductive barrier material of the first conductive barrier layer **14** has a higher electrochemical potential than copper of the first copper material portion **16** within each first bonding pad **18**. Generally, dissimilar metals and alloys have different electrode potentials due to the differences in electrochemical potentials. When two or more conductors having different electrochemical potentials come into contact in an electrolyte, such as a slurry in a chemical mechanical polishing process, the more reactive conductor acts as anode and the less reactive conductor acts as cathode for a galvanic reaction. During the first chemical mechanical polishing process, the first conductive barrier layer **14** functions as an anode, and the first copper material portion **16** functions as a cathode. Electrons flow (i.e., current flows) from the first conductive barrier layer **14** to the first copper material portion **16** within each first bonding pad **18**. The first conductive barrier layer **14** is positively charged, and surface portions of the first conductive barrier layer **14** in contact with the slurry generate positive ions of the metal within the first conductive barrier layer **14**. As a consequence, metal ions at the surface of the first conductive barrier layer **14** that contact the slurry dissociate from the first conductive barrier layer **14**, and diffuse into the slurry during the first chemical mechanical polishing process. A moat-shaped groove region **14**G, which is also referred to as a “fang region,” can be formed in an upper portion of each first conductive barrier layer **14** that is physically exposed to the slurry during the first chemical mechanical polishing process.

(66) A concave top surface of the first conductive barrier layer **14** is vertically recessed relative to a periphery of a top surface of and adjoining portion of the first copper material portion **16** within each moat-shaped groove **14**G. The moat-shaped groove **14**G is formed around the first copper material portion **16** such that the moat-shaped groove **14**G encircles the periphery of an upper region of the first copper material portion **16**. The depth of each moat-shaped groove **14**G, as measured along the vertical direction from the horizontal plane including the top surface of the first bonding-level dielectric layer to the deepest point in the moat-shaped groove **14**G, may be in a range from 1 nm to 20 nm, such as from 2 nm to 10 nm, although lesser and greater depths may also be employed.

(67) In one embodiment, a top surface of each first copper material portion **16** may be formed with a convex vertical cross-sectional profile such that a center region of the top surface protrudes farther outward from the horizontal plane including the bottom surface of the first bonding-level dielectric layer than a peripheral region of the top surface of first copper material portion **16**. The loss of material from the first conductive barrier layer **14** within each first bonding pad **18** causes the generally convex profile of the top surface of the first copper material portion **16**. In one embodiment, at least 50% of the area of the top surface of each first copper material portion **16**

may protrude outward from the horizontal plane including the top surface of the first bonding-level dielectric layer (such as the memory-level dielectric material layer **970**).

(68) Generally, a first semiconductor die **100** comprises first semiconductor devices, first metal interconnect structures embedded in first dielectric material layers and electrically connected to the first semiconductor devices, and at least one first bonding pad **18** embedded in a first bonding-level dielectric layer and electrically connected to one of the first metal interconnect structures. Each of the at least one first bonding pad **18** comprises a first conductive barrier layer **14** comprising a first conductive barrier material having a higher electrochemical potential than copper and a first copper material portion **16** laterally surrounded by the first conductive barrier layer **14**. Optionally, a suitable pre-anneal bonding process may be performed, for example, at a temperature of about 150 degrees Celsius.

(69) Referring to FIG. **2**A, a first configuration of a logic die **700** according to the first embodiment of the present disclosure is illustrated. Second-type bonding pads, which are also referred to as second bonding pads, can be subsequently formed on the logic die **700**. In this case, the logic die **700** can be referred to as a second-type semiconductor die, or as a second semiconductor die **200**. The logic die **700** includes a logic-side substrate **708**, logic-side semiconductor devices **720** overlying the logic-side substrate **708**, logic-side dielectric material layers **750** located on the logic-side semiconductor devices, and logic-side metal interconnect structures **760** embedded in the logic-side dielectric material layers **750**. In one embodiment, the logic-side substrate **708** may be a commercially available single crystal silicon wafer. In one embodiment, a plurality of second semiconductor dies **200** may be provided on the logic-side substrate **708**.

(70) Generally, the logic-side semiconductor devices may comprise any semiconductor devices, such as field effect transistors in a CMOS configuration, that may be operated in conjunction with the memory-side semiconductor devices in the memory die **900** to provide enhanced functionality. In one embodiment, the logic die **700** comprises a support circuitry (i.e., a driver/peripheral circuitry) for operation of memory devices (such as a three-dimensional array of memory elements) **920** within the memory die **900**. In one embodiment, the memory die **900** may include a three-dimensional memory device **920** including a three-dimensional array of memory elements, word lines (that may comprise a subset of the electrically conductive layers **46**), and bit lines **962**, and the logic-side semiconductor devices **720** of the logic die **700** may include a peripheral circuitry for operation of the three-dimensional array of memory elements. The peripheral circuitry may include one or more word line driver circuits that drive the word lines of the three-dimensional array of memory elements of the memory die **900**, one or more bit line driver circuits that drive the bit lines **962** of the memory die **900**, one or more word line decoder circuits that decode the addresses for the word lines, one or more bit line decoder circuits that decode the addresses for the bit lines **962**, one or more sense amplifier circuits that sense the states of memory elements within the memory opening fill structures **58** of the memory die **900**, a source power supply circuit that provides power to the horizontal semiconductor channel layer **10** in the memory die **900**, a data buffer and/or latch, and/or any other semiconductor circuit that may be used to operate three-dimensional memory device of the memory die **900**.

(71) The logic-side dielectric material layers **750** may include a dielectric material such as undoped silicate glass, a doped silicate glass, organosilicate glass, silicon nitride, a dielectric metal oxide, or a combination thereof. The logic-side dielectric material layers **750** may include one or more dielectric diffusion barrier layers (not expressly shown). In this case, each dielectric diffusion barrier layer embedded in the logic-side distal dielectric material layers **750** may include silicon carbon nitride (i.e., silicon carbonitride "SiCN", which is also referred to silicon carbide nitride), silicon nitride (Si.sub.3N.sub.4), silicon oxynitride, or any other dielectric material that is effective in blocking diffusion of copper. In one embodiment, each dielectric diffusion barrier layer embedded in the logic-side distal dielectric material layers **750** may include a dielectric material having a dielectric constant less than 5, such as SiCN having a dielectric constant of about 3.8, to

reduce RC delay of the logic-side metal interconnect structures **760**. Each dielectric diffusion barrier layer may have a thickness in a range from 20 nm to 30 nm.
(72) An optional logic-side dielectric diffusion barrier layer **752** can be formed over the logic-side dielectric material layers **750**. The logic-side dielectric diffusion barrier layer **752** can include a dielectric material that blocks copper diffusion. In one embodiment, the logic-side dielectric diffusion barrier layer **752** can include silicon nitride, silicon carbon nitride, silicon oxynitride, or a stack thereof. The thickness of the logic-side dielectric diffusion barrier layer **752** can be in a range from 5 nm to 50 nm, although lesser and greater thicknesses can also be employed.
(73) A logic-side bonding-level dielectric layer **770** can be formed over the logic-side dielectric diffusion barrier layer **752**. In case second bonding pads are subsequently formed in the logic-side bonding-level dielectric layer **770**, the logic-side bonding-level dielectric layer **770** is herein referred to as a second bonding-level dielectric layer. In one embodiment, the logic-side bonding-level dielectric layer **770** may include a dielectric material that permits dielectric-to-dielectric bonding. The logic-side bonding-level dielectric layer **770** may include, and/or consist essentially of, undoped silicate glass (i.e., silicon oxide), a doped silicate glass, organosilicate glass, silicon nitride, or a dielectric metal oxide. The thickness of the logic-side bonding-level dielectric layer **770** may be in a range from 300 nm to 3,000 nm, although lesser and greater thicknesses may also be employed. The logic-side bonding-level dielectric layer **770** may have a planar top surface.
(74) A photoresist layer (not shown) can be applied over the logic-side bonding-level dielectric layer **770**, and can be lithographically patterned to form discrete openings in each area in which second bonding pads are to be subsequently formed. In one embodiment, the second bonding pads may have shape similar to a mushroom including a lower via (e.g., stem) portion and an upper plate (e.g., cap) portion having a greater lateral dimension than the lower via portion. In this case, discrete opening in the photoresist layer can have shapes of the lower via portions of the second bonding pads. A first anisotropic etch process can be performed to transfer the pattern of the openings in the photoresist layer into an upper portion of the logic-side bonding-level dielectric layer **770**. Recess cavities can be formed within unmasked areas in the upper portion of the logic-side bonding-level dielectric layer **770**. The photoresist layer may be subsequently isotropically trimmed to increase the size of each opening in the photoresist layer. Alternatively, the photoresist layer may be removed and a new photoresist layer may be applied and patterned over the logic-side bonding-level dielectric layer **770** to form larger openings having peripheries that are laterally offset outward from a periphery of a respective recess cavity in the upper portion of the logic-side bonding-level dielectric layer **770**. A second anisotropic etch process can be performed to etch portions of the logic-side bonding-level dielectric layer **770** that are not covered by the photoresist layer. Logic-side pad cavities **779** are formed through the logic-side bonding-level dielectric layer **770** underneath discrete openings in the photoresist layer. The photoresist layer may be subsequently removed, for example, by ashing.
(75) Yet alternatively, the second bonding pads may be formed with a respective straight sidewall that vertically extend from a top surface of the logic-side bonding-level dielectric layer **770** to a bottom surface of the logic-side bonding-level dielectric layer **770**. In this case, the logic-side pad cavities **779** may be formed by applying and patterning a photoresist layer, and by removing portions of the logic-side bonding-level dielectric layer **770** by performing an anisotropic etch process. In other words, a combination of a single lithographic patterning process and a single anisotropic etch process may be employed to form the logic-side pad cavities **779** with straight sidewall(s). In one embodiment, a lateral dimension (such as a diameter or a lateral distance between facing sides of a periphery) of each opening in the photoresist layer may be in a range from 200 nm to 30 microns, such as from 400 nm to 20 microns, although lesser and greater dimensions may also be employed.
(76) A top surface of a respective logic-side metal interconnect structure **760** can be physically exposed at the bottom of each logic-side pad cavity **779**. In one embodiment, each logic-side pad

cavity **779** can have a top periphery having a horizontal cross-sectional shape of a rectangle or a rounded rectangle. A maximum lateral dimension of each logic-side pad cavity **779** may be in a range from 200 nm to 30 microns, such as from 400 nm to 20 microns, although lesser and greater dimensions may also be employed. Sidewalls of the logic-side pad cavities **779** may be vertical, or may have a taper angle greater than 0 degree and less than 30 degrees (such as a taper angle in a range from 3 degrees to 20 degrees) with respect to the vertical direction. In case second bonding pads (i.e., second-type bonding pads) are subsequently formed in the logic-side pad cavities **779**, the logic-side pad cavities **779** are herein referred to as second pad cavities.

(77) Referring to FIG. **2**B, a logic-side conductive barrier layer **784**L and a logic-side copper layer **786**L can be sequentially deposited in the logic-side pad cavities **779**. In one embodiment, the logic-side conductive barrier layer **784**L may be formed as a second-type conductive barrier layer, which is herein referred to as a second conductive barrier layer **24**L. In this case, the logic-side copper layer **786**L is formed on the second conductive barrier layer **24**L, and is herein referred to as a second copper layer **26**L.

(78) According to an aspect of the present disclosure, the second conductive barrier layer **24**L comprises, and/or consists essentially of, a second conductive barrier material having a lower electrochemical potential than copper. Non-limiting examples of metallic materials that provide a lower electrochemical potential than copper and provide suitable adhesion strength to a dielectric material (such as silicon oxide) of the second bonding-level dielectric layer (such as the logic-side bonding-level dielectric layer **770**) comprise ruthenium, titanium, or titanium nitride. In one embodiment, the second conductive barrier layer **24** consists essentially of a material selected from ruthenium, titanium, or titanium nitride. The thickness of the second conductive barrier layer **24**L (which comprises the logic-side conductive barrier layer **784**L) may be in a range from 3 nm to 30 nm, such as from 6 nm to 20 nm, although lesser and greater thicknesses can also be employed.

(79) The second copper layer **26**L (which comprises the logic-side copper layer **786**L) may consist essentially of copper. The second copper layer **26**L may be formed by depositing a copper seed layer (which is a lower surface portion of the second copper layer **26**L) employing a physical vapor deposition process, and by depositing a copper fill material layer employing electroplating. The thickness of the copper seed layer may be in a range from 20 nm to 200 nm, and the thickness of the copper fill material layer, and measured over the top surface of the second bonding-level dielectric layer, may be in a range from 300 nm to 6,000 nm, although lesser and greater thicknesses may also be employed.

(80) Referring to FIGS. **2**C and **2**D, a second chemical mechanical polishing (CMP) process can be performed to remove portions of the second copper layer **26**L (comprising the logic-side copper layer **786**L) and the second conductive barrier layer **24**L (comprising the logic-side conductive barrier layer **784**L) from above the top surface of the second bonding-level dielectric layer (comprising the logic-side bonding-level dielectric layer **770**). Each remaining portion of the second copper layer **26**L comprises the second copper material portion **26**. Each remaining portion of the second conductive barrier layer **24**L comprises a second conductive barrier layer **24**. Each second copper material portion **26** is a logic-side copper material portion **786**. Each second conductive barrier layer **24** is a logic-side conductive barrier layer **784** embedded within the logic-side bonding dielectric layer **770**. Each contiguous combination of a second conductive barrier layer **24** and a second copper material portion **26** constitutes a second bonding pad **28**, which in this embodiment is a logic-side bonding pad **788**.

(81) Generally, second bonding pads **28** embedded in a second bonding-level dielectric layer (such as the logic-side bonding-level dielectric layer **770**) and electrically connected to a respective one of the second metal interconnect structures (such as the logic-side metal interconnect structures **760**) are formed. Each of the second bonding pads **28** comprises a second conductive barrier layer **24** including and/or consisting essentially of a second conductive barrier material having a lower electrochemical potential than copper. Each of the second bonding pads **28** comprises a second

copper material portion **26** laterally surrounded by the second conductive barrier layer **24**.
(82) The second chemical mechanical polishing process proceeds under a condition in which the second conductive barrier material of the second conductive barrier layer **24** has a lower electrochemical potential than copper of the second copper material portion **26** within each second bonding pad **28**. During the second chemical mechanical polishing process of the present disclosure, the second conductive barrier layer **26** functions as a cathode, and the second copper material portion **26** functions as an anode. Electrons flow (i.e., current flows) from the second copper material portion **26** to the second conductive barrier layer **24** within each first bonding pad **18**. The second copper material portion **26** is positively charged, and surface portions of the second copper portion **26** in contact with the slurry generate positive copper ions. As a consequence, the copper ions at the surface of the second copper material portion **26** that contact the slurry dissociate from the second copper portion **26**, and diffuse into the slurry during the second chemical mechanical polishing process, which results in dishing. Dishing leads to a concave upper surface in the second copper portion **26**.
(83) In one embodiment, the top surface of each second conductive barrier layer **24** may be formed within a horizontal plane including the top surface of the second bonding-level dielectric layer (which comprises the logic-side bonding-level dielectric layer **770**). In one embodiment, each second conductive barrier layer **24** may comprise a respective top surface having an inner periphery and an outer periphery within the horizontal plane including the top surface of the second bonding-level dielectric layer.
(84) In one embodiment, the entirety or a major portion of each second copper material portion **26** may be located at or below the horizontal plane including the top surface of the second bonding-level dielectric layer. In one embodiment, the middle of each second copper material portion **26** is recessed farther than its edges. Thus, for each second bonding pad **28** located within a respective second pad cavity, a vertical distance between a concave top surface of the second copper material portion **26** and the horizontal plane increases with a lateral distance from a respective most proximal point within a periphery of the pad cavity that is located within the horizontal plane, i.e., with a lateral distance from a top periphery of the second pad cavity. In other words, the farther a point within a concave top surface of a second copper material portion **26** is laterally from the periphery of the second pad cavity, the farther the point is vertically from the horizontal plane.
(85) The maximum dishing depth, i.e., the maximum distance between the concave top surface of each second copper material portion **26** from the horizontal plane including the top surface of the second bonding-level dielectric layer, may be in a range from 1 nm to 30 nm, such as from 2 nm to 10 nm, although lesser and greater maximum dishing depths may also be employed. In one embodiment, at least 50%, such as from 80% to 100%, of the area of the top surface of each second copper material portion **26** may be recessed below the horizontal plane including the top surface of the second bonding-level dielectric layer (which comprises the logic-level dielectric material layer **770**).
(86) Generally, the second semiconductor die **200** comprises second semiconductor devices, second metal interconnect structures embedded in second dielectric material layers and electrically connected to the second semiconductor devices, and at least one second bonding pad **28** embedded in a second bonding-level dielectric layer and electrically connected to one of the second metal interconnect structures.
(87) The second bonding pad **28** comprises a second conductive barrier layer **24** comprising a second conductive barrier material having a lower electrochemical potential than copper and a second copper material portion **26** laterally surrounded by the second conductive barrier layer **24**. Optionally, a suitable pre-anneal bonding process may be performed, for example, at a temperature of about 150 degrees Celsius.
(88) Referring to FIG. **3**A, the first substrate including the plurality of first semiconductor dies **100** and the second substrate including the plurality of second semiconductor dies **200** can be aligned to

each other for bonding. In this case, the first bonding pads **18** and the second bonding pads **28** can face each other. In the illustrated example, the first semiconductor dies **100** can be memory dies **900**, the second semiconductor dies **200** can be logic dies **700**, the first bonding pads **18** can be memory-side bonding pads **988**, and the second bonding pads **28** can be logic-side bonding pads **788**.

(89) The first bonding-level dielectric layer is brought into contact with the second bonding-level dielectric layer while each mating pair of a first bonding pad **18** and a second bonding pad **28** remains aligned to each other. Thus, the convex first bonding pads **18** having respective convex first copper material portions **16** are aligned to respective concave second bonding pads having respective concave second copper material portions **26**. A cavity, which is herein referred to as a bonding cavity, may be formed between each mating pair of a first bonding pad **18** and a second bonding pad **28**.

(90) In one embodiment, the volume of each bonding cavity may be in a range from 0.01% to 0.5%, such as from 0.03% 0.3% of the combined volume of the mating pair of a first copper material portion **16** and a second copper material portion **26**. Generally, each cavity may include a volume of a moat-shaped groove **14**G and a volume between a convex surface of a first copper material portion **16** and a concave surface of a second copper material portion **26**.

(91) Referring to FIG. **3**B, an anneal process can be performed at an elevated temperature to induce copper-to-copper bonding between each mating pair of a first bonding pad **18** and a second bonding pad **28**. The elevated temperature of the anneal process may be in a range from 300 degrees Celsius to 450 degrees Celsius, such as from 350 degrees Celsius to 425 degrees Celsius.

(92) The first copper material portions **16** and the second copper material portions **26** expand in volume during the anneal process, and are bonded to a respective mating copper material portion. In one embodiment, a wafer-to-wafer bonding process may be performed to bond an array of first semiconductor dies **100** on the first substrate **108** to an array of second semiconductor dies **200** on the second substrate **208**.

(93) Generally, copper-to-copper bonding can be induced between the second copper material portions **26** and the first copper material portions **16** such that each second bonding pad **28** can be bonded to a respective one of the first bonding pad **18**, and each first bonding pad **18** can be bonded to a respective one of the second bonding pads **28**. In one embodiment, at least 80%, such from 85% to 100%, and/or from 90% to 99%, of a bonding interface between the first bonding pad **18** and the second bonding pad **28** within each bonded pair of a first bonding pad **18** and a second bonding pad **28** may be formed on a side of the second semiconductor die **200** relative to a horizontal plane including an interface between the first bonding dielectric layer and the second bonding dielectric layer after bonding the second bonding pads **28** to the first bonding pad **18**.

(94) In one embodiment, a bonding interface between the first bonding pads **18** and the second bonding pads **28** between each bonded pair of a first bonding pad **18** and a second bonding pad **28** comprises a non-planar (i.e., curved) interface at which a convex surface of the first copper material portion **16** contacts a concave surface of the second copper material portion **26**. In one embodiment, the entirety of a periphery of the convex surface of the first copper material portion **16** in a bonded pair of a first bonding pad **18** and a second bonding pad **28** may be adjoined to a surface of the first bonding-level dielectric layer (**970** or **770**).

(95) In one embodiment, the first conductive barrier layer **14** in a bonded pair of a first bonding pad **18** and a second bonding pad **28** may be vertically spaced from the second bonding pad **28** by a peripheral portion of the first copper material portion **16** that contacts a sidewall of the first bonding-level dielectric layer (which comprises the memory-side bonding-level dielectric layer **970** in the illustrated example). The peripheral portion of the first copper material portion **16** may fill the volume of the moat-shaped groove **14**G in the first conductive barrier layer **14**. Thus, the bonding pads **18** and **28** of the first embodiment form a good bond, even in the case of partial lateral misalignment.

(96) In contrast, in prior art bonded assemblies containing opposing convex bonding pads surrounded by respective moat-shaped grooves, there may be insufficient contact between the bonding pads in case of the partial lateral misalignment of the opposing bonding pads. In other words, if only the moat-shaped grooves overlap each other, then there may not be sufficient copper filling of both grooves to form a sufficient bond. Likewise, in other prior art bonded assemblies containing opposing concave bonding pads, there may be insufficient copper filling in the middle of the bonding pads which forms a void in the middle of the bonding interface between opposing convex bonding pads. The void degrades the quality of the bond.
(97) In one embodiment, the first conductive barrier layer **14** comprises a vertically-extending portion contacting a sidewall of the first bonding-level dielectric layer (such as the memory-side bonding-level dielectric layer **970**) and having a contoured top surface that continuously extends around the first copper material portion **16** and having a concave vertical cross-sectional profile (i.e., the vertical cross-sectional profile of a mot-shaped groove **14**G). In one embodiment, a point of the concave top surface laterally midway between an inner sidewall and an outer sidewall of the concave top surface is more distal from a horizontal plane at which the first bonding-level dielectric layer contacts the second bonding-level dielectric layer than a point of the inner sidewall that is mot proximal to the horizontal plane, and than a point of the outer sidewall that is most proximal to the horizontal plane. This is due to the "U-shaped" vertical cross-sectional profile of the moat-shaped groove **14**G overlying the annular top surface of the first conductive barrier layer **14**.
(98) In one embodiment, a vertically extending portion of the first conductive barrier layer **14** in a bonded pair of a first bonding pad **18** and a second bonding pad **28** comprises a moat-shaped groove **14**G that laterally surrounds the first copper material portion **16** and having a concave profile. A peripheral portion of the first copper material portion **16** fills the moat-shaped groove **14**G, and is disposed between the first conductive barrier layer **14** and the second bonding pad **28**.
(99) In one embodiment, at least 80%, such as from 85% to 100%, and/or from 90% to 99%, of the bonding interface between the first bonding pads **18** and the second bonding pads **28** may be located on a side of the second semiconductor die **200** with respect to a horizontal plane including an interface between the first bonding-level dielectric layer and the second bonding-level dielectric layer.
(100) In one embodiment, the first conductive barrier layer **14** in a bonded pair of a first bonding pad **18** and a second bonding pad **28** is vertically spaced from the second bonding pad **28** by the first copper material portion **16**, and the second conductive barrier layer **24** is in direct contact with the first copper material portion **16**.
(101) In one embodiment, the first conductive barrier layer **14** consists essentially of a material selected from cobalt, tantalum, or tantalum nitride. In one embodiment, the second conductive barrier layer **24** consists essentially of a material selected from ruthenium, titanium, or titanium nitride.
(102) In one embodiment, the second bonding-level dielectric layer is in direct contact with the first bonding-level dielectric layer at a horizontal plane, and a horizontal surface of a vertically-extending portion of the second conductive barrier layer **24** is located within the horizontal plane. In one embodiment, an annular surface of a vertically-extending portion of the first conductive barrier layer **14** that faces the second semiconductor die **200** and is most proximal to the second semiconductor die **200** is vertically spaced from the horizontal plane by a peripheral portion of the first copper material portion **16**, i.e., by a sub-portion of the first copper material portion **16** that fills a moat-shaped groove **14**G. In one embodiment, the annular surface comprises a non-planar grooved surface having a concave vertical cross-sectional profile, and the non-planar grooved surface is contacted entirely by the first copper material portion **16**.
(103) Referring to FIG. **3**C, the memory-side substrate **908** may optionally be thinned from the backside by grinding, polishing, an anisotropic etch, or an isotropic etch. The thinning process can

continue until horizontal portions of the through-substrate liners **386** are removed, and horizontal surfaces of the through-substrate via structures (if present) **388** are physically exposed. Generally, end surfaces of the through-substrate via structures **388** can be physically exposed by thinning the backside of the memory-side substrate **908**, which may be the substrate of a memory die. The thickness of the memory-side substrate **908** after thinning may be in a range from 1 micron to 30 microns, such as from 2 microns to 15 microns, although lesser and greater thicknesses can also be employed.
(104) An optional backside insulating layer **930** may be formed on the backside of the memory-side substrate **908**. The backside insulating layer **930** includes an insulating material such as silicon oxide. The thickness of the backside insulating layer **930** can be in a range from 50 nm to 500 nm, although lesser and greater thicknesses can also be employed. A photoresist layer (not shown) may be applied over the backside insulating layer **930**, and may be lithographically patterned to form opening over areas of the through-substrate via structures **388**. An etch process can be performed to form via cavities through the backside insulating layer **930** underneath each opening in the photoresist layer. A top surface of a through-substrate via structure **388** can be physically exposed at the bottom of each via cavity through the backside insulating layer **930**.
(105) At least optional one metallic material can be deposited into the openings through the backside insulating layer **930** and over the planar surface of the backside insulating layer **930** to form a metallic material layer. The at least one metallic material can include copper, aluminum, ruthenium, cobalt, molybdenum, and/or any other metallic material that may be deposited by physical vapor deposition, chemical vapor deposition, electroplating, vacuum evaporation, or other deposition methods. For example, a metallic nitride liner material (such as TiN, TaN, or WN) may be deposited directly on the physically exposed surfaces of the through-substrate via structures **388**, on sidewalls of the openings through the backside insulating layer **930**, and over the physically exposed planar surface of the backside insulating layer **930**. The thickness of the metallic nitride liner material can be in a range from 10 nm to 100 nm, although lesser and greater thicknesses can also be employed. At least one metallic fill material such as copper or aluminum can be deposited over the metallic nitride liner material. In one embodiment, the at least one metallic fill material can include a stack of a high-electrical-conductivity metal layer (such as a copper layer or an aluminum layer) and an underbump metallization (UBM) layer stack for bonding a solder ball thereupon. Exemplary UBM layer stacks include, but are not limited to, an Al/Ni/Au stack, an Al/Ni/Cu stack, a Cu/Ni/Au stack, a Cu/Ni/Pd stack, a Ti/Ni/Au stack, a Ti/Cu/Ni/Au stack, a Ti—W/Cu stack, a Cr/Cu stack, and a Cr/Cu/Ni stack. The thickness of the metallic material layer over the planar horizontal surface of the backside insulating layer **930** can be in a range from 0.5 microns to 10 microns, such as from 1 micron to 5 microns, although lesser and greater thicknesses can also be employed.
(106) The at least one metallic fill material and the metallic material layer can be subsequently patterned to form discrete backside bonding pads **936** contacting a respective one of the through-substrate via structures **388**. The backside bonding pads **936** can function as external bonding pads that can be employed to electrically connect various nodes of within the memory die **900** and the logic die **700** to external nodes, such as bonding pads on a packaging substrate or C4 bonding pads of another semiconductor die. For example, solder material portions **938** can be formed on the backside bonding pads **936**, and a C4 bonding process or a wire bonding process can be performed to electrically connect the backside bonding pads **936** to external electrically active nodes.
(107) Generally, backside bonding pads **936** can be formed on a backside surface of the memory die **900** (which may be a memory die) that is located on an opposite side of the bonding interfaces with the logic die **700**. Through-substrate via structures **388** can vertically extend through the memory die **900**, and can provide electrical connection between the backside bonding pads **936** and a subset of the bonding pads (**18**, **28**).
(108) In an alternative aspect of the first embodiment shown in FIG. **3**D, the horizontal

semiconductor channel layer **10**, the through-substrate liners **386**, the through-substrate via structures **388**, the through-memory-level dielectric liner **486** and/or the through-memory-level via structure **488** may be omitted. In this alternative configuration, the entire first substrate **108** is removed rather than thinned to expose the tips of the memory opening fill structures **58**. After removing the first substrate **108**, the memory films at the exposed tips of the memory opening fill structures **58** are removed by selective etching to expose the tips of the semiconductor channels in the memory opening fill structures **58**. A doped semiconductor source layer **110** is formed in contact with the exposed tips of the semiconductor channels. A conductive source line (e.g., source contact) **112** is then formed on the doped semiconductor source layer **110**.

(109) Referring to FIG. **4**A, a second configuration of the first exemplary structure is illustrated. The second configuration of the first exemplary structure can be derived from the first configuration of the first exemplary structure illustrated in FIG. **3**A by forming the memory-side bonding pads **988** as second bonding pads **28**, and by forming the logic-side bonding pads **788** as first bonding pads **18**. In other words, the memory-side bonding pads **988** are formed as combinations of a respective second conductive barrier layer **24** and a respective second copper material portion **28** employing the second chemical mechanical planarization process described above, and the logic-side bonding pads **788** are formed as combinations of a respective first conductive barrier layer **14** and a respective first copper material portion **18** employing the first chemical mechanical planarization process described above.

(110) Each of the first conductive barrier layers **14** in the second configuration of the first exemplary structure may be formed with the same material composition and with the same geometrical features as the first conductive barrier layers **14** in the first configuration of the first exemplary structure as described above. Each of the first copper material portions **16** in the second configuration of the first exemplary structure may be formed with the same material composition and with the same geometrical features as the first copper material portion **16** in the first configuration of the first exemplary structure as described above. Each of the second conductive barrier layers **24** in the second configuration of the first exemplary structure may be formed with the same material composition and with the same geometrical features as the second conductive barrier layers **24** in the first configuration of the first exemplary structure as described above. Each of the second copper material portions **26** in the second configuration of the first exemplary structure may be formed with the same material composition and with the same geometrical features as the second copper material portions **26** in the first configuration of the first exemplary structure as described above.

(111) Referring to FIG. **4**B, the processing steps described with reference to FIG. **3**B may be performed to induce copper-to-copper bonding between each mating pair of a respective first copper material portion **16** and a respective second copper material portion **26**. Dielectric-to-dielectric bonding (such as silicon oxide-to-silicon oxide bonding) may be performed simultaneously with the copper-to-copper bonding between the first bonding-level dielectric layer and the second bonding-level dielectric layer. The bonded pairs of a respective first copper material portion **16** and a respective second copper material portion **26** may have the same geometry as in the first configuration of the first exemplary structure described with reference to FIG. **3**B.

(112) Referring to FIGS. **4**C and **4**D, the processing steps described with reference to respective FIGS. **3**C and **3**D may be performed as described above.

(113) Referring to FIG. **5**, a third configuration of a logic die **700** is illustrated in which the size of opposing bonding pads differs from each other. Generally, the third configuration of the logic die **700** can be derived from the first configuration of the logic die **700** that is illustrated with reference to FIGS. **2**C and **2**D by increasing the size of each second bonding pad **28**. Specifically, the lateral dimensions of the second bonding pads **28** can be selected such that the area of the top surface of each second copper material portion **26** is designed to include the entirety of the area of the top surface a respective first bonding pad **18** to be subsequently bonded thereto. Thus, the periphery of

the concave top surface of each second bonding pad **28** may have a shape in a plan view that is laterally offset outward from a mirror image of the periphery of the outer sidewall of the first conductive barrier layer **14** of the first bonding pad **18** to which the second bonding pad **28** is to be subsequently bonded.

(114) Referring to FIG. **6**A, the processing steps of FIG. **3**A can be performed with any needed changes in view of the increase in the size of the second bonding pads **28**, such that the second bonding pads **28** face the first bonding pads **18**. In one embodiment, the annular top surfaces of the second conductive barrier layers **24** of the second bonding pads **28** may directly contact the top surface of the first bonding-level dielectric layer (such as the memory-side bonding-level dielectric layer **970**). In one embodiment, an entirety of a top surface of the second conductive barrier layer **26** of a second bonding pad **28** may contact a horizontal surface of the first bonding-level dielectric layer.

(115) Referring to FIG. **6**B, the processing steps of FIG. **3**B can be performed to induce metal-to-metal bonding between each mating pair of a first copper material portion **16** and a second coper material portion **26**. Dielectric-to-dielectric bonding (such as silicon oxide-to-silicon oxide bonding) may be performed simultaneously with the copper-to-copper bonding between the first bonding-level dielectric layer and the second bonding-level dielectric layer. Generally, the bonded pairs of a respective first copper material portion **16** and a respective second copper material portion **26** may have the same geometry, or similar geometry, as in the first configuration of the first exemplary structure described with reference to FIG. **3**B. In the third configuration of the first exemplary structure, the entirety of a bonding surface of the first copper material portion **16** in a bonded pair of a first bonding pad **18** and a second bonding pad **28** may be in contact with a bonding surface of the second copper material portion **26**. In one embodiment, an annular peripheral surface of the second copper material portion **26** of a second bonding pad **28** may be in contact with the first bonding-level dielectric layer (which may comprise a memory-side bonding-level dielectric layer **970**). In one embodiment, an entirety of a top surface of the second conductive barrier layer **24** in a second bonding pad **28** may be in contact with the first bonding-level dielectric layer.

(116) In this configuration, the convex first bonding pads **18** have a smaller width than the concave second bonding pads **28**, such that the edges of the second copper material portions **26** fill the grooves **14**G. This improves the bonding quality between the opposing bonding pads. Subsequently, the processing steps of FIG. **3**C or **3**D may be performed to thin the backside of the memory-side substrate **908** (or to remove it completely) and to form various backside bonding structures or a combination of the doped semiconductor source layer **110** and source line **112**.

(117) Referring to FIG. **7**A, a fourth configuration of the first exemplary structure is illustrated. The fourth configuration of the first exemplary structure can be derived from the third configuration of the first exemplary structure illustrated in FIG. **6**A by forming the memory-side bonding pads **988** as second bonding pads **28**, and by forming the logic-side bonding pads **788** as first bonding pads **18**. In other words, the memory-side bonding pads **988** are formed as combinations of a respective second conductive barrier layer **24** and a respective second copper material portion **28** employing the second chemical mechanical planarization process described above, and the logic-side bonding pads **788** are formed as combinations of a respective first conductive barrier layer **14** and a respective first copper material portion **18** employing the first chemical mechanical planarization process described above.

(118) Referring to FIG. **7**B, the processing steps described with reference to FIG. **6**B may be performed to induce copper-to-copper bonding between each mating pair of a respective first copper material portion **16** and a respective second copper material portion **26**. Dielectric-to-dielectric bonding (such as silicon oxide-to-silicon oxide bonding) may be performed simultaneously with the copper-to-copper bonding between the first bonding-level dielectric layer and the second bonding-level dielectric layer. The bonded pairs of a respective first copper material

portion **16** and a respective second copper material portion **26** may have the same geometry as in the third configuration of the first exemplary structure described with reference to FIG. **6**B. Subsequently, the processing steps described with reference to FIGS. **3**C and **3**D may be performed to thin the backside of the memory-side substrate **908** (or to remove it completely) and to form various backside bonding structures or a combination of the doped semiconductor source layer **110** and source line **112**.

(119) Referring to the first embodiment, and all drawings associated with the first exemplary structure, a bonded assembly comprises a first semiconductor die **100** comprising first semiconductor devices (**920** or **720**) and a first bonding pad **18**. The first bonding pad comprises a first copper material portion **16** and a first conductive barrier layer **14** comprising a first conductive barrier material having a higher electrochemical potential than copper located between the first semiconductor devices (**920** or **720**) and the first copper material portion **16**. The bonded assembly also includes a second semiconductor die **200** comprising second semiconductor devices (**720** or **920**) and a second bonding pad **28**. The second bonding pad comprises a second copper material portion **26** and a second conductive barrier layer **24** comprising a second conductive barrier material having a lower electrochemical potential than copper located between the second semiconductor devices (**720** or **920**) and the second copper material portion **26**. The second bonding pad is bonded to the first bonding pad.

(120) In one embodiment, a bonding interface between the first bonding pads **18** and the second bonding pads **28** comprises a non-planar interface at which a convex surface of the first copper material portion **16** contacts a concave surface of the second copper material portion **26**.

(121) In one embodiment, the first semiconductor die **100** further comprises at least one first dielectric material layer, first metal interconnect structures embedded in the at least one first dielectric material layer and electrically connected to the first semiconductor devices, and a first bonding-level dielectric layer, wherein the first bonding pad is embedded in the first bonding-level dielectric layer and electrically connected to at least one of the first metal interconnect structures. The second semiconductor die further comprises at least one second dielectric material layer, second metal interconnect structures embedded in the at least one second dielectric material layer and electrically connected to the second semiconductor devices, and a second bonding-level dielectric layer, wherein the second bonding pad is embedded in the second bonding-level dielectric layer and electrically connected to at least one of the second metal interconnect structures.

(122) In one embodiment, an entirety of a periphery of the convex surface of the first copper material portion **16** is adjoined to a surface of the first bonding-level dielectric layer (**970** or **770**). In one embodiment, the first conductive barrier layer **14** is vertically spaced from the second bonding pad **28** by a peripheral portion of the first copper material portion **16** that contacts a sidewall of the first bonding-level dielectric layer (**970** or **770**).

(123) In one embodiment, the first conductive barrier layer **14** comprises a vertically-extending portion contacting a sidewall of the first bonding-level dielectric layer (**970** or **770**) and having a contoured top surface that continuously extends around the first copper material portion **16** and having a concave vertical cross-sectional profile; and a point of the concave top surface laterally midway between an inner sidewall and an outer sidewall of the concave top surface is more distal from a horizontal plane at which the first bonding-level dielectric layer (**970** or **770**) contacts the second bonding-level dielectric layer (**770** or **970**) than a point of the inner sidewall that is mot proximal to the horizontal plane, and than a point of the outer sidewall that is most proximal to the horizontal plane.

(124) In one embodiment, a vertically extending portion of the first conductive barrier layer **14** comprises a moat-shaped groove **14**G that laterally surrounds the first copper material portion **16** and having a concave profile; and a peripheral portion of the first copper material portion **16** fills the moat-shaped groove **14**G and is disposed between the first conductive barrier layer **14** and the

second bonding pad **28**. In one embodiment, at least 80% of the bonding interface between the first bonding pads **18** and the second bonding pads **28** is located on a side of the second semiconductor die **200** with respect to a horizontal plane including an interface between the first bonding-level dielectric layer (**970** or **770**) and the second bonding-level dielectric layer (**770** or **970**).
(125) In one embodiment, the first conductive barrier layer **14** is vertically spaced from the second bonding pad **28** by the first copper material portion **16**, and the second conductive barrier layer **24** is in direct contact with the first copper material portion **16**. In one embodiment, an entirety of a bonding surface of the first copper material portion **16** is in contact with a bonding surface of the second copper material portion **26**. In one embodiment, an annular peripheral surface of the second copper material portion **26** is in contact with the first bonding-level dielectric layer (**970** or **770**), and an entirety of a top surface of the second conductive barrier layer **24** is in contact with the first bonding-level dielectric layer (**970** or **770**).
(126) In one embodiment, the first conductive barrier layer **14** consists essentially of a material selected from cobalt, tantalum, or tantalum nitride. In one embodiment, the second conductive barrier layer **24** consists essentially of a material selected from ruthenium, titanium, or titanium nitride.
(127) In one embodiment, the second bonding-level dielectric layer (**770** or **970**) is in direct contact with the first bonding-level dielectric layer (**970** or **770**) at a horizontal plane; a horizontal surface of a vertically-extending portion of the second conductive barrier layer **24** is located within the horizontal plane; and an annular surface of a vertically-extending portion of the first conductive barrier layer **14** that faces the second semiconductor die **200** and is most proximal to the second semiconductor die **200** is vertically spaced from the horizontal plane by a peripheral portion of the first copper material portion **16**. In one embodiment, the annular surface comprises a non-planar grooved surface having a concave vertical cross-sectional profile, and the non-planar grooved surface is contacted entirely by the first copper material portion **16**.
(128) The combination of a convex bonding surface of a first copper material portion **16** and a concave bonding surface of a second copper material portion **26** reduces the volume of a cavity that needs to be filled through volume expansion of the copper material portions (**16**, **26**) during a bonding anneal process. By reducing the volume of cavities to be filled during the bonding anneal process, the copper-to-copper bonding between mating pairs of copper material portions (**16**, **26**) can be formed with greater adhesion strength and with less defects, thereby increasing the bonding yield.
(129) Referring to FIG. **8**A, a first configuration of a memory die **900** according to a second embodiment of the present disclosure is illustrated. The memory die **900** can be employed to form first-type bonding pads, or first bonding pads thereupon. In one embodiment, the memory-side substrate **908** may be a portion of a commercially available silicon wafer having a thickness in a range from 500 microns to 1 mm. In one embodiment, a plurality of first semiconductor dies **100** can be provided on a first wafer.
(130) The first configuration of the memory die **900** in the second embodiment of the present disclosure may be derived from the first configuration of the memory die **900** in the first embodiment of the present disclosure illustrated in FIG. **1**A by forming a first conductive barrier layer **13**L and a first copper layer **15**L. The first conductive barrier layer **13**L and the first copper layer **15**L are employed to form first bonding pads in subsequent processing steps. The first conductive barrier layer **13**L may comprise a memory-side conductive barrier layer **984**L. The first copper layer **15**L may comprise a memory-side copper layer **986**L.
(131) According to an aspect of the present disclosure, the material composition and the thickness of the first conductive barrier layer **13**L are selected such that at least 10% of the volume fraction of a copper material portion in each first bonding pad may include (200) copper grains, where (200) is the Miller Index of the plane of the copper grains. The at least 10% of the volume fraction may be formed in polycrystalline copper after an anneal at an elevated temperature, such as an

anneal at 150 degrees Celsius, for example. As used herein, crystallographic orientations of copper grains in a copper layer refer to the crystallographic plane within the copper layer that generates a Bragg peak in an x-ray scan in which the incidence angle and the reflection angle are the same, which is typically referred to as a θ:2θ scan. The incidence angle is measured between the direction of the incident X-ray beam and the horizontal plane that is perpendicular to the growth direction of the copper layer. The reflection angle is measured between the direction of the reflected X-ray beam and the horizontal plane that is perpendicular to the growth direction of the copper layer. In a θ:2θ scan, the detector angle 2θ (as measured between the direction of the incident beam and the reflected beam) is double the incidence angle θ that is measured between the incident beam and the sample surface, such as a horizontally-extending portion of the top surface of the first copper layer **15**L that is parallel to the top surface of the first bonding-level dielectric layer. Thus, (200) copper grains are copper grains that generate a (200) peak in a θ:2θ X-ray scan. Similarly, (110) copper grains are copper grains that generate a (110) peak in a θ:2θ X-ray scan. (111) copper grains are copper grains that generate (111) peak in a θ:2θ X-ray scan. Generally, the (200) copper grains grow along a direction that is perpendicular to the [100] crystallographic orientation direction of copper, and therefore, include (100) crystallographic planes as horizontal planes. The (110) copper grains grow along a direction that is perpendicular to the [110] crystallographic orientation direction of copper, and therefore, include (110) crystallographic planes as horizontal planes. The (111) copper grains grow along a direction that is perpendicular to the [111] crystallographic orientation direction of copper, and therefore, include (111) crystallographic planes as horizontal planes.

(132) Generally, slow recrystallization of electroplated copper results in formation of a predominant fraction of copper grains as (111) copper grains. In contrast, formation of (200) copper grains by annealing electroplated copper at any significant volume fraction is more difficult to achieve. According to an aspect of the present disclosure, the material composition and the thickness of the first conductive barrier layer **13**L can be selected such that each first copper pad may include (200) copper grains at a volume fraction not less than 10% after an anneal process at 150 degrees Celsius prior to a bonding process. The present inventors found material compositions and/or thickness ranges that result formation of (200) copper grains at a volume fraction of at least 10% within first bonding pads that are formed after a first chemical mechanical planarization process, and preferably by performing a subsequent anneal at an elevated temperature, such as 100 to 200 degrees Celsius, such as 150 degrees Celsius.

(133) Specifically, in one embodiment, the first conductive barrier layer **13**L comprises a titanium layer **13**A that is formed directly on, and thus is in direct contact with the first bonding-level dielectric layer (which may comprise the memory-side bonding-level dielectric layer **970**). The first conductive barrier layer **13**L has a thickness of at least 10 nm. For example, the titanium layer **13**A is in direct contact with the first copper material portion **15** and has a thickness greater than 15 nm. The thickness of the titanium layer **13**A may be in a range from 16 nm to 30 nm.

(134) In a first example, which is herein referred to as Option 1, the first conductive barrier layer **13**L may consist of the titanium layer **13**A having a thickness greater than 15 nm, such as a thickness in a range from 16 nm to 30 nm. In this case, the first copper layer **15**L can be formed directly on the titanium layer **13**A, which is the entirety of the first conductive barrier layer **13**L.

(135) In a second example, which is herein referred to as Option 2, the first conductive barrier layer **13**L comprises a tantalum layer **13**B that is formed between the titanium layer **13**A and the first copper layer **15**L. In this case, the first copper layer **15**L can be formed directly on the tantalum layer **13**B. The thickness of the first conductive barrier layer **13**L may be in a range from 10 nm to 60 nm, the thickness of the titanium layer **13**A may be in a range from 5 nm to 30 nm, and the thickness of the tantalum layer **13**B may be in a range from 5 nm to 30 nm, such as from 5 nm to 20 nm.

(136) The thickness of the first copper layer **15**L, as measured over a horizontal top surface of the

first bonding-level dielectric layer, may be in a range from 300 nm to 6 microns, such as from 600 nm to 3 microns, although lesser and greater thicknesses may also be employed. The first copper layer **15**L as deposited may be polycrystalline, amorphous or substantially amorphous, i.e., may include crystallographic grains at a volume fraction less than 10%, such as less than 5%, and/or less than 2%. The first copper layer **15**L may be deposited using a two step deposition process which includes first depositing a copper seed layer by physical vapor deposition (e.g., sputtering) followed by depositing a copper fill layer on the copper seed layer by plating (e.g., electroplating).
(137) Referring to FIG. **8**B, a first chemical mechanical polishing (CMP) process can be performed to remove portions of the first copper layer **15**L (which may comprise the memory-side copper layer **986**L) and the first conductive barrier layer **13**L (which may comprise the memory-side conductive barrier layer **984**L) from above the top surface of the first bonding-level dielectric layer (which may comprise the memory-side bonding-level dielectric layer **970**). Each remaining portion of the first copper layer **15**L comprises the first copper material portion **15**. Each remaining portion of the first conductive barrier layer **13**L comprises a first conductive barrier layer **13**. Each first copper material portion **15** is a memory-side copper material portion **986**. Each first conductive barrier layer **13**L is a memory-side conductive barrier layer **984** embedded within the memory-side bonding dielectric layer **970**. Each contiguous combination of a first conductive barrier layer **13** and a first copper material portion **15** constitutes a first bonding pad **17**, which is a memory-side bonding pad **988**.
(138) An anneal at an elevated temperature may be performed to induce formation of crystalline grains within the first copper material portions **15**. Generally, the crystallization characteristics of thin copper films, such as copper films having a thickness less than 3,000 nm and/or less than 1,000 nm, is different from crystallization characteristics of thick copper films, and generates more (111) grains than thick copper films. According to an aspect of the present disclosure, the material composition and the thickness of the first conductive barrier layers **13** are selected such that more than 10% of the volume fraction of all copper material within each first copper material portion **15** includes (200) copper grains after a low temperature anneal at a relatively low temperature, such as an anneal at a temperature of 150 degrees Celsius. The duration of the anneal at the elevated temperature may be in a range from 10 minutes to 600 minutes, such as from 30 minutes to 300 minutes, although lesser and greater durations may also be employed. As discussed above, formation of more than 10% of the volume fraction of all copper material within each first copper material portion **15** as (200) copper grains may be achieved under several conditions. For example, the first conductive barrier layer **13** in each first bonding pad **17** may comprises a titanium layer **13**A in direct contact with the first bonding-level dielectric layer. In one example, the titanium layer **13**A in each first bonding pad **17** may be in direct contact with the first copper material portion **15** and may have a thickness greater than 15 nm. In this example, the first copper material portion **15** in contact with the titanium layer **13**A may have a concave top (i.e., bonding) surface, as discussed above with respect to the first embodiment. In another example, the first conductive barrier layer **13** in each first bonding pad **17** further comprises a tantalum layer **13**B disposed between the titanium layer **13**A and the first copper material portion **15**. In this example, the first copper material portion **15** in contact with the tantalum layer **13**B may have a convex top (i.e., bonding) surface, as discussed above with respect to the first embodiment.
(139) First bonding pads **17** are embedded in a first bonding-level dielectric layer (such as the memory-side bonding-level dielectric layer **970**) and electrically connected to a respective one of the first metal interconnect structures (such as the memory-side metal interconnect structures **960**). Each of the first bonding pads **17** comprises a first conductive barrier layer **13** including a material composition or a material stack that induces subsequent formation of more than 10% in volume fraction of (200) copper grains in the first copper material portions **15** after a subsequent anneal process. Each of the first bonding pads **17** comprises a first copper material portion **15** at least partially laterally surrounded by the first conductive barrier layer **13**. A top surface of the first

copper portion **15** in each first bonding pad **17** may be vertically recessed relative to the horizontal plane including the top surface of the first bonding-level dielectric layer.

(140) Generally, a first semiconductor die **100** comprises first semiconductor devices, first metal interconnect structures embedded in first dielectric material layers and electrically connected to the first semiconductor devices, and a first bonding pad **17** embedded in a first bonding-level dielectric layer and electrically connected to one of the first metal interconnect structures. The first bonding pad **17** comprises a first conductive barrier layer **13** and a first copper material portion **15** laterally surrounded by the first conductive barrier layer **13** and including (200) copper grains at a volume fraction not less than 10%.

(141) Referring to FIG. **9**A, a first configuration of a logic die **700** for formation of a second exemplary structure is illustrated. Second-type bonding pads, which are also referred to as second bonding pads, can be subsequently formed on the logic die **700**. In this case, the logic die **700** can be referred to as a second-type semiconductor die, or as a second semiconductor die **200**. The logic die **700** includes a logic-side substrate **708**, logic-side semiconductor devices **720** overlying the logic-side substrate **708**, logic-side dielectric material layers **750** located on the logic-side semiconductor devices, and logic-side metal interconnect structures **760** embedded in the logic-side dielectric material layers **750**. In one embodiment, the logic-side substrate **708** may be a commercially available silicon wafer having a thickness in a range from 500 microns to 2 mm. In one embodiment, a plurality of second semiconductor dies **200** may be provided on a second wafer.

(142) The first configuration of the logic die **700** in the second embodiment of the present disclosure may be derived from the first configuration of the logic die **700** in the first embodiment of the present disclosure illustrated in FIG. **2**A by forming a second conductive barrier layer **23**L and a second copper layer **25**L. The second conductive barrier layer **23**L and the second copper layer **25**L are employed to form second bonding pads in subsequent processing steps. The second conductive barrier layer **23**L may be embodied as a logic-side conductive barrier layer **784**L. The second copper layer **25**L may be embodied as a logic-side copper layer **786**L.

(143) According to an aspect of the present disclosure, the material composition and/or the thickness of the second conductive barrier layer **23**L are selected such that at least 10% of the volume fraction of a copper material portion in each second bonding pad may include (200) copper grains upon an anneal at an elevated temperature. Specifically, the second conductive barrier layer **23**L comprises a titanium layer **23**A that is formed directly on, and thus is in direct contact with the second bonding-level dielectric layer (which may comprise the logic-side bonding-level dielectric layer **770**). The thickness of the second conductive barrier layer **23**L is at least 10 nm. The titanium layer **23**A is in direct contact with the second copper material portion **25** and has a thickness greater than 15 nm. The thickness of the titanium layer **23**A may be in a range from 16 nm to 30 nm.

(144) In a first example, which is herein referred to as Option I, the second conductive barrier layer **23**L may consist of the titanium layer **23**A having a thickness greater than 15 nm, such as a thickness in a range from 16 nm to 30 nm. In this case, the second copper layer **25**L can be formed directly on the titanium layer **23**A, which constitutes the entirety of the second conductive barrier layer **23**L.

(145) In a second example, which is herein referred to as Option II, the second conductive barrier layer **23**L comprises a tantalum layer **23**B that is formed between the titanium layer **23**A and the second copper layer **25**L. The thickness of the second conductive barrier layer **23** may be in a range from 10 nm to 60 nm, the thickness of the titanium layer **23**A may be in a range from 5 nm to 30 nm, and the thickness of the tantalum layer **23**B may be in a range from 5 nm to 30 nm, such as from 5 nm to 20 nm.

(146) The thickness of the second copper layer **25**L, as measured over a horizontal top surface of the second bonding-level dielectric layer, may be in a range from 300 nm to 6 microns, such as from 600 nm to 3 microns, although lesser and greater thicknesses may also be employed. The second copper layer **25**L as deposited may be polycrystalline, amorphous, or substantially

amorphous.

(147) Referring to FIG. **9**B, a second chemical mechanical polishing (CMP) process can be performed to remove portions of the second copper layer **25**L (comprising the logic-side copper layer **786**L) and the second conductive barrier layer **23**L (comprising the logic-side conductive barrier layer **784**L) from above the top surface of the second bonding-level dielectric layer (comprising the logic-side bonding-level dielectric layer **770**). Each remaining portion of the second copper layer **25**L comprises the second copper material portion **25**. Each remaining portion of the second conductive barrier layer **23**L comprises a second conductive barrier layer **23**. Each second copper material portion **25** is a logic-side copper material portion **786**. Each second conductive barrier layer **23** is a logic-side conductive barrier layer **784** embedded within the logic-side bonding dielectric layer **770**. Each contiguous combination of a second conductive barrier layer **23** and a second copper material portion **25** constitutes a second bonding pad **27**, which is a logic-side bonding pad **788**.

(148) An anneal at an elevated temperature may be performed to induce formation of crystal grains within the second copper material portions **25**. As described above, the material composition and/or the thickness of the second conductive barrier layers **23** are selected such that at least 10% of the volume fraction of all copper material within each second copper material portion **25** includes (200) copper grains after a low temperature anneal described above. In one example, the second conductive barrier layer **23** in each second bonding pad **27** may comprises a titanium layer **23**A in direct contact with the second bonding-level dielectric layer. The titanium layer **23**A in each second bonding pad **27** may be in direct contact with the second copper material portion **25** and may have a thickness greater than 15 nm. In another example, the second conductive barrier layer **23** in each second bonding pad **27** comprises a tantalum layer **23**B disposed between the titanium layer **23**A and the second copper material portion **25**.

(149) Second bonding pads **27** are embedded in a second bonding-level dielectric layer (such as the logic-side bonding-level dielectric layer **770**) and electrically connected to a respective one of the second metal interconnect structures (such as the logic-side metal interconnect structures **760**). Each of the second bonding pads **27** comprises a second conductive barrier layer **23** including a material composition or a material stack that induces subsequent formation of at least 10% in volume fraction of (200) copper grains in the second copper material portions **25** upon a subsequent anneal process. Each of the second bonding pads **27** comprises a second copper material portion **25** at least partially laterally surrounded by the second conductive barrier layer **23**. A top surface of the second copper portion **25** in each second bonding pad **27** may be vertically recessed relative to the horizontal plane including the top surface of the second bonding-level dielectric layer.

(150) Generally, a second semiconductor die **200** comprises second semiconductor devices, second metal interconnect structures embedded in second dielectric material layers and electrically connected to the second semiconductor devices, and a second bonding pad **27** embedded in a second bonding-level dielectric layer and electrically connected to one of the second metal interconnect structures. The second bonding pad **27** comprises a second conductive barrier layer **23** and a second copper material portion **25** laterally surrounded by the second conductive barrier layer **23** and including (200) copper grains at a volume fraction of at least 10%.

(151) Referring to FIG. **10**A, the first wafer including the plurality of first semiconductor dies **100** and the second wafer including the plurality of second semiconductor dies **200** can be aligned to each other for bonding. In this case, the front side of each wafer including the first bonding pads **17** or the second bonding pads **27** can face each other. The second bonding pads **27** can be arranged in a mirror image pattern of the first bonding pads **17**, and the first wafer and the second wafer can be aligned such that each second bonding pad faces a respective first bonding pad and vice versa. In the illustrated example, the first semiconductor dies **100** can be memory dies **900**, the second semiconductor dies **200** can be logic dies **700**, the first bonding pads **17** can be memory-side

bonding pads **988**, and the second bonding pads **27** can be logic-side bonding pads **788**.
(152) In one configuration in which the first and the second embodiments are combined one of the first bonding pad **17** or the second bonding pad **27** has a concave vertical cross-sectional profile, and the other one of the first bonding pad **17** or the second bonding pad **27** has a convex vertical cross-sectional profile. Specifically, one of the first bonding pad **17** or the second bonding pad **27** may have the Option I configuration where the copper material portion directly contacts the titanium layer to form the concave vertical cross-sectional profile. The other one of the first bonding pad **17** or the second bonding pad **27** may have the Option II configuration where the copper material portion directly contacts the tantalum layer of the titanium-tantalum layer stack to form the convex vertical cross-sectional profile. The bonding of concave bonding pad to the opposing convex bonding pad may form the non-planar bonding interface and improve the bonding quality, as described above.
(153) The first and second dies are brought into contact with each other such that each mating pair of a first bonding pad **17** and a second bonding pad **27** are aligned to each other. A cavity, which is herein referred to as a bonding cavity, may be formed between each mating pair of a first bonding pad **17** and a second bonding pad **27**. In one embodiment, the volume of each bonding cavity may be in a range from 0.01% to 0.5%, such as from 0.03% 0.3% of the combined volume of the mating pair of a first copper material portion **15** and a second copper material portion **25**.
(154) Referring to FIG. **10**B, a thermal anneal process can be performed at an elevated temperature to induce copper-to-copper bonding between each mating pair of a first bonding pad **17** and a second bonding pad **27**. The elevated temperature of the thermal anneal process may be in a range from 300 degrees Celsius to 450 degrees Celsius, such as from 350 degrees Celsius to 425 degrees Celsius. The first copper material portions **15** and the second copper material portions **25** expand in volume during the thermal anneal process, and contact and are bonded to a respective mating copper material portion. In one embodiment, a bonding process may be performed to bond an array of first semiconductor dies **100** on a first substrate to an array of second semiconductor dies **200** on a second substrate.
(155) Generally, copper-to-copper bonding can be induced between the second copper material portions **25** and the first copper material portions **15** such that each second bonding pad **27** can be bonded to a respective one of the first bonding pad **17**, and each first bonding pad **17** can be bonded to a respective one of the second bonding pads **27**. According to an aspect of the present disclosure, the first copper material portions **15** and the second copper material portions **25** include (200) copper grains at a volume fraction of at least 10%, such as from 10% to 80% and/or from 15% to 70%, and/or from 20% to 60%. The (200) copper grains provide a greater linear volume expansion along the grain growth direction (i.e., along the vertical direction perpendicular to the bonding interface) than directions that are perpendicular to the grain growth direction. In other words, the volume expansion of copper grains is not isotropic, but is the greatest along the growth directions of the (200) copper grains. For example, the ratio of the volume expansion rate (per unit temperature change) along the grain growth direction (i.e., a vertical direction) for (200) copper grains is believed to be greater than the volume expansion rate (per unit temperature change) along the grain growth direction for (110) and (111) copper grains.
(156) Referring back to FIG. **10**B, the higher the volume fraction of (200) copper grains in the first copper material portion **15** and in the second copper material portion **25**, the greater the volume expansion of the first copper material portions **15** and the second copper material portions **25**, and the lower the probability of insufficient contact between mating pairs of a respective first copper material portions **15** and the second copper material portions **25**. By increasing the volume fraction of (200) copper grains above 10%, embodiments of the present disclosure increase copper volume expansion for the purpose of copper-to-copper bonding along the direction that is substantially perpendicular to the bonding interfaces, and improves the quality of copper-to-copper bonding.
(157) Generally, the second bonding pads **27** can be bonded to the first bonding pads **17** by

inducing copper-to-copper bonding between the second copper material portions **25** and the first copper material portions **15**. In one embodiment, the second copper material portion includes (200) copper grains at a volume fraction of at least 10% (such as a volume fraction in a range from 10% to 80% and/or from 15% to 70%, and/or from 20% to 60%) prior to and after bonding the second bonding pads **27** to the first bonding pads **17**. In one embodiment, the first conductive barrier layer **13** in each bonded pair of a first bonding pad **17** and a second bonding pad **27** includes a layer stack including a titanium layer **13**A and a tantalum layer **13**B that contacts the first copper material portion **15**, or a titanium layer **13**A having a thickness not less than 15 nm. In one embodiment, the second conductive barrier layer **23** includes a layer stack including a titanium layer **23**A and a tantalum layer **23**B that contacts the second copper material portion **25**, or a titanium layer **23**A having a thickness not less than 15 nm.

(158) Referring to FIG. **10**C, the processing steps of FIG. **3**C can be performed to thin the backside of the memory-side substrate **908**, and to form various backside bonding structures.

(159) Alternatively, referring to FIG. **10**D, a doped semiconductor source layer **110** is formed in contact with the exposed tips of the semiconductor channels, and a conductive source line (e.g., source contact) **112** is then formed on the doped semiconductor source layer **110**, as described with regard to FIG. **3**D.

(160) Referring to FIG. **11**A, a second configuration of a logic die **700** according to a second embodiment of the present disclosure is illustrated. The second configuration of the logic die **700** can be derived from the first configuration of the logic die **700** illustrated in FIG. **10**A by employing a second conductive barrier layer **33**L having a different material composition than the second conductive barrier layer **23**L in the first configuration of the logic die **700** in the second embodiment of the present disclosure as illustrated in FIG. **9**A. The second conductive barrier layer **33**L is formed as a logic-side conductive barrier layer **784**L. A second copper layer **35**L can be formed on the second conductive barrier layer **33**L. The second copper layer **35**L is formed as a logic-side copper layer **786**L.

(161) According to an aspect of the present disclosure, the material composition and the thickness of the second conductive barrier layer **33**L are selected such that at least 95% of the volume fraction of a copper material portion in each second bonding pad may include (111) copper grains, such as after an anneal at an elevated temperature, such as a temperature of 100 to 200 degrees Celsius, for example 150 degrees Celsius. In some embodiments, the second conductive barrier layer **33**L comprises a tantalum layer **33**A in direct contact with the second bonding-level dielectric layer (such as the logic-side bonding-level dielectric layer **770**) without contacting an underlying titanium layer. In some embodiments, the second conductive barrier layer **33**L comprises a titanium layer **33**B having a thickness of 10 nm or less in direct contact with the second copper material portion **35**. In some embodiments, the second conductive barrier layer **33**L is a stack of an underlying tantalum layer **33**A and an overlying titanium layer **33**B having a thickness of 10 nm or less in direct contact with the second copper material portion **35**.

(162) In a first example, which is herein referred to as Option A, the second conductive barrier layer **33**L consists of a tantalum layer **33**A having a thickness greater than 10 nm, such as a thickness in a range from 15 nm to 30 nm. In this case, the second copper layer **35**L can be formed directly on the tantalum layer **33**A, which comprises the entirety of the second conductive barrier layer **23**L.

(163) In a second example, which is herein referred to as Option B, the second conductive barrier layer **23**L comprises a stack of a tantalum layer **33**A and a titanium layer **33**B. The tantalum layer **33**A is in direct contact with the second bonding-level dielectric layer (such as the logic-side bonding-level dielectric layer **770**). The titanium layer **33**B is in direct contact with the second copper material portion **35**. The thickness of the titanium layer **33**B is less than the thickness of the tantalum layer **33**A. The thickness of the second conductive barrier layer **33**L may be in a range from 15 nm to 30 nm, the thickness of the tantalum layer **33**A may be in a range from 15 nm to 30

nm, and the thickness of the titanium layer **33**B may be in a range from 0 nm to 10 nm, such as from 1 nm to 6 nm and/or from 2 nm to 4 nm.

(164) In a third example, which is herein referred to as Option C, the second conductive barrier layer **33**L consists of a titanium layer having a thickness of 10 nm or less, such as from 1 nm to 6 nm and/or from 2 nm to 4 nm. In this case, the second copper layer **35**L can be formed directly on the titanium layer **33**B, which comprises the entirety of the second conductive barrier layer **23**L.

(165) The thickness of the second copper layer **35**L, as measured over a horizontal top surface of the second bonding-level dielectric layer, may be in a range from 300 nm to 6 microns, such as from 600 nm to 3 microns, although lesser and greater thicknesses may also be employed. The second copper layer **35**L as deposited may be polycrystalline, amorphous, or substantially amorphous.

(166) Referring to FIG. **11**B, a second chemical mechanical polishing (CMP) process can be performed to remove portions of the second copper layer **35**L (comprising the logic-side copper layer **786**L) and the second conductive barrier layer **33**L (comprising the logic-side conductive barrier layer **784**L) from above the top surface of the second bonding-level dielectric layer (comprising the logic-side bonding-level dielectric layer **770**). Each remaining portion of the second copper layer **35**L comprises the second copper material portion **35**. Each remaining portion of the second conductive barrier layer **33**L comprises a second conductive barrier layer **33**. Each second copper material portion **35** is a logic-side copper material portion **786**. Each second conductive barrier layer **33** is a logic-side conductive barrier layer **784** embedded within the logic-side bonding dielectric layer **770**. Each contiguous combination of a second conductive barrier layer **33** and a second copper material portion **35** constitutes a second bonding pad **37**, which is a logic-side bonding pad **788**.

(167) An anneal at an elevated temperature may be performed to induce formation of crystal grains within the second copper material portions **35**. According to an aspect of the present disclosure, the material composition and/or the thickness of the second conductive barrier layers **33** are selected such that at least 95% of the volume fraction of all copper material within each second copper material portion **35** includes (111) copper grains. The duration of the anneal at the elevated temperature of 100 to 200 degrees Celsius may be in a range from 30 minutes to 600 minutes, such as from 30 minutes to 300 minutes, although lesser and greater durations may also be employed. Without wishing to be bound by a particular theory, the present inventors believe that copper formed directly on thin titanium barrier layers (e.g., having a thickness or 10 or less) or directly on tantalum barrier layers without an underlying titanium barrier layer have a predominant (111) orientation in which at least 95 volume percent of all grains have a (111) orientation.

(168) Second bonding pads **37** are embedded in a second bonding-level dielectric layer (such as the logic-side bonding-level dielectric layer **770**) and electrically connected to a respective one of the second metal interconnect structures (such as the logic-side metal interconnect structures **760**). Each of the second bonding pads **37** comprises a second conductive barrier layer **33** including a material composition or a material stack that induces subsequent formation of at least 95% in volume fraction of (111) copper grains in the second copper material portions **35** upon a subsequent anneal process. Thus, at least 95%, such as from 95% to 100%, and/or from 96% to 99%, of the volume fraction of a copper material portion in each second bonding pad **35** may include (111) copper grains upon an anneal at an elevated temperature (which may be about 150 degrees Celsius). Each of the second bonding pads **37** comprises a second copper material portion **35** at least partially laterally surrounded by the second conductive barrier layer **33**. A top surface of the second copper portion **35** in each second bonding pad **37** may be vertically recessed relative to the horizontal plane including the top surface of the second bonding-level dielectric layer.

(169) Generally, a second semiconductor die **200** comprises second semiconductor devices, second metal interconnect structures embedded in second dielectric material layers and electrically connected to the second semiconductor devices, and a second bonding pad **37** embedded in a

second bonding-level dielectric layer and electrically connected to one of the second metal interconnect structures. The second bonding pad **37** comprises a second conductive barrier layer **33** and a second copper material portion **35** laterally surrounded by the second conductive barrier layer **33** and including (111) copper grains at a volume fraction of at least 95%.

(170) Referring to FIG. **12**A, a second configuration of the second exemplary structure according to the second embodiment of the present disclosure is illustrated. A first wafer can be provided, which includes a plurality of first semiconductor dies **100** that includes multiple instances of the first semiconductor die **100** as illustrated in FIG. **8**C. A second wafer can be provided, which includes a plurality of second semiconductor dies **200** that includes multiple instances of the second semiconductor die as illustrated in FIG. **11**B. The first wafer and the second wafer can be aligned to each other for bonding. In this case, the front side of each wafer including the first bonding pads **17** or the second bonding pads **37** can face each other. The second bonding pads **37** can be arranged in a mirror image pattern of the first bonding pads **17**, and the first wafer and the second wafer can be aligned such that each second bonding pad faces a respective first bonding pad and vice versa. In the illustrated example, the first semiconductor dies **100** can be memory dies **900**, the second semiconductor dies **200** can be logic dies **700**, the first bonding pads **17** can be memory-side bonding pads **988**, and the second bonding pads **37** can be logic-side bonding pads **788**.

(171) The first bonding-level dielectric layer can be brought into contact with the second bonding-level dielectric layer while each mating pair of a first bonding pad **17** and a second bonding pad **37** remains aligned to each other. A cavity, which is herein referred to as a bonding cavity, may be formed between each mating pair of a first bonding pad **17** and a second bonding pad **37**. In one embodiment, the volume of each bonding cavity may be in a range from 0.01% to 0.5%, such as from 0.03% 0.3% of the combined volume of the mating pair of a first copper material portion **15** and a second copper material portion **35**.

(172) Referring to FIG. **12**B, a thermal anneal process can be performed at an elevated temperature to induce copper-to-copper bonding between each mating pair of a first bonding pad **17** and a second bonding pad **37**. The elevated temperature of the thermal anneal process may be in a range from 300 degrees Celsius to 450 degrees Celsius, such as from 350 degrees Celsius to 425 degrees Celsius. The first copper material portions **15** and the second copper material portions **35** expand in volume during the thermal anneal process, and contact and are bonded to a respective mating copper material portion.

(173) Generally, copper-to-copper bonding can be induced between the second copper material portions **35** and the first copper material portions **15** such that each second bonding pad **37** can be bonded to a respective one of the first bonding pad **17**, and each first bonding pad **17** can be bonded to a respective one of the second bonding pads **37**. According to an aspect of the present disclosure, the first copper material portions **15** include (200) copper grains at a volume fraction of at least 10%, such as from 10% to 80% and/or from 15% to 70%, and/or from 20% to 60%. According to another aspect of the present disclosure, the second copper material portions **35** include (111) copper grains at a volume fraction of at least 95%, such as from 95% to 100%, and/or from 96% to 99%.

(174) The (200) copper grains provide a greater liner volume expansion along the grain growth direction (i.e., along the vertical direction) than directions that are perpendicular to the grain growth direction as discussed above. The (111) copper grain provide greater surface diffusivity than the (200) copper grains. The higher surface diffusivity of the (111) copper grains promote copper diffusion along the bonding interfaces between the first copper material portions **15** and the second copper material portions **35** to enhance the adhesion at the bonding interface during the bonding process.

(175) Generally, the second bonding pads **37** can be bonded to the first bonding pads **17** by inducing copper-to-copper bonding between the second copper material portions **35** and the first

copper material portions **15**. In one embodiment, the first conductive barrier layer **13** in each bonded pair of a first bonding pad **17** and a second bonding pad **37** includes a layer stack including a titanium layer **13**A and a tantalum layer **13**B that contacts the first copper material portion **15**; or a titanium layer **13**A having a thickness not less than 15 nm. In one embodiment, the second copper material portion **35** in each bonded pair of a first bonding pad **17** and a second bonding pad **37** includes (111) copper grains at a volume fraction not less than 95% prior to and after bonding the second bonding pad **37** to the first bonding pad **17**. In one embodiment, the second conductive barrier layer **33** in each bonded pair of a first bonding pad **17** and a second bonding pad **37** consists of: a layer stack including a tantalum layer **33**A and a titanium layer **33**B that contacts the second copper material portion **35**; a titanium layer **33**B having a thickness less than 10 nm; or a tantalum layer **33**A without an underlying titanium layer.

(176) Referring to FIG. **12**C, the processing steps of FIG. **3**C can be performed to thin the backside of the memory-side substrate **908**, and to form various backside bonding structures. Alternatively, a doped semiconductor source layer **110** is formed in contact with the exposed tips of the semiconductor channels, and a conductive source line (e.g., source contact) **112** is then formed on the doped semiconductor source layer **110**, as described with regard to FIGS. **3**D and **10**D.

(177) Referring to FIG. **13**A, a third configuration of a memory die **900** according to a second embodiment of the present disclosure is illustrated. The third configuration of the memory **900** may be derived from the first configuration of the memory die **900** illustrated in FIG. **8**A by employing the second conductive barrier layer **33**L in lieu of the first conductive barrier layer **13**L. The second conductive barrier layer **33**L is formed as a memory-side conductive barrier layer **984**L. A second copper layer **35**L can be formed on the second conductive barrier layer **33**L. The second copper layer **35**L is formed as a memory-side copper layer **986**L.

(178) According to an aspect of the present disclosure, the material composition and the thickness of the second conductive barrier layer **33**L are selected such that at least 95% of the volume fraction of a copper material portion in each second bonding pad may include (111) copper grains, such as after an anneal at an elevated temperature, for example at 150 degrees Celsius. The second conductive barrier layer **33**L may have the same material composition and the same thickness range as discussed above.

(179) Referring to FIG. **13**B, a second chemical mechanical polishing (CMP) process can be performed to remove portions of the second copper layer **35**L (comprising the memory-side copper layer **986**L) and the second conductive barrier layer **33**L (comprising the memory-side conductive barrier layer **984**L) from above the top surface of the second bonding-level dielectric layer (comprising the memory-side bonding-level dielectric layer **970**). Each remaining portion of the second copper layer **35**L comprises the second copper material portion **35**. Each remaining portion of the second conductive barrier layer **33**L comprises a second conductive barrier layer **33**. Each second copper material portion **35** is a memory-side copper material portion **986**. Each second conductive barrier layer **33** is a memory-side conductive barrier layer **984** embedded within the memory-side bonding dielectric layer **970**. Each contiguous combination of a second conductive barrier layer **33** and a second copper material portion **35** constitutes a second bonding pad **37**, which is a memory-side bonding pad **988**.

(180) An anneal at an elevated temperature may be performed to induce formation of crystal grains within the second copper material portions **35**. According to an aspect of the present disclosure, the material composition and/or the thickness of the second conductive barrier layers **33** are selected such at least 95% of the volume fraction of all copper material within each second copper material portion **35** includes (111) copper grains, such as after a low temperature anneal at a relatively low temperature, such as an anneal at a temperature of 150 degrees Celsius, as described above.

(181) Second bonding pads **37** are embedded in a second bonding-level dielectric layer (such as the memory-side bonding-level dielectric layer **970**) and electrically connected to a respective one of the second metal interconnect structures (such as the memory-side metal interconnect structures

960). Each of the second bonding pads **37** comprises a second conductive barrier layer **33** including a material composition or a material stack that induces subsequent formation of at least 95% in volume fraction of (111) copper grains in the second copper material portions **35**. Thus, at least 95%, such as from 95% to 100%, and/or from 96% to 99% of the volume fraction of a copper material portion in each second bonding pad **35** may include (111) copper grains. Each of the second bonding pads **37** comprises a second copper material portion **35** laterally surrounded by the second conductive barrier layer **33**. A top surface of the second copper portion **35** in each second bonding pad **37** may be vertically recessed relative to the horizontal plane including the top surface of the second bonding-level dielectric layer.
(182) Generally, the second semiconductor die **200** comprises second semiconductor devices, second metal interconnect structures embedded in second dielectric material layers and electrically connected to the second semiconductor devices, and a second bonding pad **37** embedded in a second bonding-level dielectric layer and electrically connected to one of the second metal interconnect structures. The second bonding pad **37** comprises a second conductive barrier layer **33** and a second copper material portion **35** at least partially laterally surrounded by the second conductive barrier layer **33** and including (111) copper grains at a volume fraction of at least 95%.
(183) Referring to FIG. **14**A, a third configuration of the second exemplary structure according to the second embodiment of the present disclosure is illustrated. A first wafer can be provided, which includes a plurality of first semiconductor dies **100**. The first semiconductor die **100** may be the same as the logic **700** illustrated in FIG. **9**B. A second wafer can be provided, which includes a plurality of second semiconductor dies **200** that includes multiple instances of the memory die **900** illustrated in FIG. **13**B. The first wafer and the second wafer can be aligned to each other for bonding. In this case, the bonding pads **27** (which may be the same as bonding pad **17**) and **37** can face each other. The first bonding-level dielectric layer can be brought into contact with the second bonding-level dielectric layer while each mating pair of bonding pads **27** and **37** remains aligned to each other.
(184) Referring to FIG. **14**B, the processing steps of FIG. **12**B can be performed to induce bonding between the bonding pads **27** and **37**, as described above.
(185) Referring to FIG. **14**C, the processing steps of FIG. **3**C can be performed to thin the backside of the memory-side substrate **908**, and to form various backside bonding structures. Alternatively, a doped semiconductor source layer **110** is formed in contact with the exposed tips of the semiconductor channels, and a conductive source line (e.g., source contact) **112** is then formed on the doped semiconductor source layer **110**, as described with regard to FIGS. **3**D and **10**D.
(186) Referring to the first, second and third configurations of the second exemplary structure and FIGS. **8**A to **14**C related thereto, a bonded assembly comprises a first semiconductor die **100** comprising first semiconductor devices (**920** or **720**) and a first bonding pad **17**. The first bonding pad comprises a first copper material portion **15** containing (200) copper grains at a volume fraction of at least 10% and a first conductive barrier layer **13** located between the first semiconductor devices and the first copper material portion. The bonded assembly also includes a second semiconductor die **200** comprising second semiconductor devices (**720** or **920**) and a second bonding pad (**27** or **37**). The second bonding pad comprises a second copper material portion (**25** or **35**) and a second conductive barrier layer (**23** or **33**) located between the second semiconductor devices and the second copper material portion. The second bonding pad is bonded to the first bonding pad.
(187) In one embodiment, the first conductive barrier layer **13** comprises a titanium layer **13**A having a thickness greater than 15 nm. In another embodiment, the first conductive barrier layer **13** comprises a stack of a tantalum layer **13**B and a titanium layer **13**A, wherein the tantalum layer is disposed between the titanium layer and the first copper material portion **15**.
(188) In the first configuration of the second embodiment illustrated in FIGS. **8**A to **10**D, the second copper material portion **25** contains (200) copper grains at a volume fraction of at least

10%. In one embodiment, the second conductive barrier layer **23** comprises a titanium layer **23**A having a thickness greater than 15 nm. In another embodiment, the second conductive barrier layer **23** comprises a stack of a tantalum layer **23**B and a titanium layer **23**A, wherein the tantalum layer is disposed between the titanium layer and the first copper material portion **25**.

(189) In the second and third configurations of the second embodiment illustrated in FIGS. **9**A to **14**C, the second copper material portion **35** contains (111) copper grains at a volume fraction of at least 95%. In one embodiment, the second conductive barrier layer **33** comprises a tantalum layer **33**A in direct contact with the second copper material portion **35**. In another embodiment, the second conductive barrier layer **33** comprises a titanium layer **33**A having a thickness not greater than 10 nm in direct contact with the second copper material portion **35**. In another embodiment, the second conductive barrier layer **33** comprises a stack of a tantalum layer **33**B and a titanium layer **33**A, wherein the titanium layer is disposed between the tantalum layer and the first copper material portion **35**.

(190) It is believed that the (200) copper grains provide higher mean expansion distance per unit temperature change (e.g., per degree Celsius) than the (110) copper grains and the (111) copper grains. In contrast, it is also believed that the (111) copper grains provide higher surface diffusivity than (110) copper grains and (200) copper grains. Embodiments utilizing at least 10% in volume fraction of (200) copper grains in a first copper material portion **15** and a second copper material portion **25** utilize increased volume expansion of the (200) copper grains along the vertical direction (along the direction perpendicular to the bonding interfaces) to provide enhanced copper-to-copper bonding between copper material portions (**15**, **25**). Embodiments utilizing at least 10% in volume fraction of (200) copper grains in a first copper material portion **15** and at least 95% in volume fraction of (111) copper grains in a second copper material portion **35** utilize the combination of increased volume expansion from the first copper material portion **15** and enhanced surface diffusivity provided by the second copper material portion **35** to provide enhanced copper-to-copper bonding between copper material portions (**15**, **35**).

(191) Although the foregoing refers to particular embodiments, it will be understood that the disclosure is not so limited. It will occur to those of ordinary skill in the art that various modifications may be made to the disclosed embodiments and that such modifications are intended to be within the scope of the disclosure. Compatibility is presumed among all embodiments that are not alternatives of one another. The word "comprise" or "include" contemplates all embodiments in which the word "consist essentially of" or the word "consists of" replaces the word "comprise" or "include," unless explicitly stated otherwise. Where an embodiment using a particular structure and/or configuration is illustrated in the present disclosure, it is understood that the present disclosure may be practiced with any other compatible structures and/or configurations that are functionally equivalent provided that such substitutions are not explicitly forbidden or otherwise known to be impossible to one of ordinary skill in the art. All of the publications, patent applications and patents cited herein are incorporated herein by reference in their entirety.

Claims

1. A bonded assembly, comprising: a first semiconductor die comprising first semiconductor devices and a first bonding pad, wherein the first bonding pad comprises a first copper material portion and a first conductive barrier layer comprising a first conductive barrier material having a higher electrochemical potential than copper located between the first semiconductor devices and the first copper material portion; and a second semiconductor die comprising second semiconductor devices and a second bonding pad, wherein the second bonding pad comprises a second copper material portion and a second conductive barrier layer comprising a second conductive barrier material having a lower electrochemical potential than copper located between the second semiconductor devices and the second copper material portion; wherein: the second bonding pad is

bonded to the first bonding pad; a bonding interface between the first bonding pads and the second bonding pads comprises a non-planar interface at which a convex surface of the first copper material portion contacts a concave surface of the second copper material portion; the first semiconductor die further comprises: at least one first dielectric material layer; first metal interconnect structures embedded in the at least one first dielectric material layer and electrically connected to the first semiconductor devices; and a first bonding-level dielectric layer, wherein the first bonding pad is embedded in the first bonding-level dielectric layer and electrically connected to at least one of the first metal interconnect structures; and the second semiconductor die further comprises: at least one second dielectric material layer; second metal interconnect structures embedded in the at least one second dielectric material layer and electrically connected to the second semiconductor devices; and a second bonding-level dielectric layer, wherein the second bonding pad is embedded in the second bonding-level dielectric layer and electrically connected to at least one of the second metal interconnect structures; an entirety of a periphery of the convex surface of the first copper material portion is adjoined to a surface of the first bonding-level dielectric layer; and the first conductive barrier layer is vertically spaced from the second bonding pad by a peripheral portion of the first copper material portion that contacts a sidewall of the first bonding-level dielectric layer.

2. A bonded assembly, comprising: a first semiconductor die comprising first semiconductor devices and a first bonding pad, wherein the first bonding pad comprises a first copper material portion and a first conductive barrier layer comprising a first conductive barrier material having a higher electrochemical potential than copper located between the first semiconductor devices and the first copper material portion; and a second semiconductor die comprising second semiconductor devices and a second bonding pad, wherein the second bonding pad comprises a second copper material portion and a second conductive barrier layer comprising a second conductive barrier material having a lower electrochemical potential than copper located between the second semiconductor devices and the second copper material portion; wherein: the second bonding pad is bonded to the first bonding pad; wherein a bonding interface between the first bonding pads and the second bonding pads comprises a non-planar interface at which a convex surface of the first copper material portion contacts a concave surface of the second copper material portion; the first semiconductor die further comprises: at least one first dielectric material layer; first metal interconnect structures embedded in the at least one first dielectric material layer and electrically connected to the first semiconductor devices; and a first bonding-level dielectric layer, wherein the first bonding pad is embedded in the first bonding-level dielectric layer and electrically connected to at least one of the first metal interconnect structures; and the second semiconductor die further comprises: at least one second dielectric material layer; second metal interconnect structures embedded in the at least one second dielectric material layer and electrically connected to the second semiconductor devices; and a second bonding-level dielectric layer, wherein the second bonding pad is embedded in the second bonding-level dielectric layer and electrically connected to at least one of the second metal interconnect structures; the second bonding-level dielectric layer is in direct contact with the first bonding-level dielectric layer at a horizontal plane; a horizontal surface of a vertically-extending portion of the second conductive barrier layer is located within the horizontal plane; and an annular surface of a vertically-extending portion of the first conductive barrier layer that faces the second semiconductor die and is most proximal to the second semiconductor die is vertically spaced from the horizontal plane by a peripheral portion of the first copper material portion.

3. The bonded assembly of claim 2, wherein the annular surface comprises a non-planar grooved surface having a concave vertical cross-sectional profile, and the non-planar grooved surface is contacted entirely by the first copper material portion.